IMPACT 360®

System Monitoring, Messaging, and Alarms Guide

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About This Guide

This guide provides a printed list of all alarM that can be generated by the system, including the corrective action for each alarm.

Intended Audience

This guide is designed to be used by:

- Verint and Business Partner professional services staff responsible for the configuration, operation, or maintenance of systems.
- Customers' IT staff responsible for system maintenance.
- Enterprise administrators and system integrators

Related Documents

The following documents are referenced in this document:

- *Enterprise Manager Configuration and Administration Guide*

The following documents are related to this document:

- *Enterprise Manager Configuration and Administration Guide*
- *Call Lock Configuration and User Guide*
- *Recorder Configuration and Administration Guide*
- *Archive Configuration and Administration Guide*
- *Framework Configuration and Administration Guide*

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Chapter 1

Introduction

This book explains how to view and configure alarms and also provides lists of all the alarms that can be raised by the Impact 360 system, including the summaries and corrective actions for each alarm.

This book includes the following chapters:

- **Chapter 2 "System Monitor and Alarms"** - Explains how to view and configure alarms. The chapter discusses topics such as viewing alarm summaries, viewing corrective actions for alarms, acknowledging alarms, monitoring alarm activity on specific servers, configuring global alarm settings, configuring individual alarm settings, and working with alarm notification profiles.
- **Chapter 3 "Working with SNMP MIB Files"** - Explains how to get the SNMP MIB files and provides an example of compiling these files to work with your system.
- **Chapter 4 "Understanding the Alarm Spreadsheet" -** Discusses the *Alarms Spreadsheet*. The *Alarms Spreadsheet* is an Excel document that lists all alarms the system can raise. The default value of each attribute of each alarm is also listed in this spreadsheet. The *Alarms Spreadsheet* is included as an attachment to the PDF version of this document, and is also available as a separate standalone document.
- **Chapter 5 "Troubleshooting"** - Provides troubleshooting information related to alarms.

Chapter 2

System Monitor and Alarms

Alarms alert you to conditions on the managed servers of which you should be aware, including conditions that can adversely affect server performance. Use the System Monitor to view alarm summaries, to view and configure specific alarms, and to view managed server communication errors.

You can view alarm summaries at the Enterprise, Site Group, Site, and Server nodes.

From the alarm summaries, you can drill down to view detailed information on specific alarms that will help you determine the corrective action for the alarm. After you perform the corrective action, you acknowledge the alarm to move it into alarm history and place it in an inactive state.

You can assign alarm notification profiles to each alarm. An alarm notification profile ensures that email notifications are sent to specific people and SNMP traps are sent to specific SNMP nodes when that alarm is triggered.

NOTE Specific alarms on individual managed servers are handled by the Enterprise Manager Agent (EMA) application installed on each managed server by default. The EMA communicates with, and obtains alarm configuration information from, Enterprise Manager.

You can also launch the Enterprise Manager Agent (EMA) application or the Recorder Manager (RM) application on a managed server from the Enterprise Manager and use the System Monitor on the EMA or RM to view detailed status information about the managed server.

Information is presented in the following topics:

- View Alarm Summaries, page 9
- View Active Alarms and Acknowledge Alarms, page 16
- Search for Servers with Alarms, page 25
- Configure Alarms, page 29
- Work with Alarm Notification Profiles, page 47
- View Detailed Status of a Managed Server, page 60

View Alarm Summaries

You can view an Alarm Summary at each node of the Installations tree. For example, you can view a summary of all alarms in the Enterprise, all alarms in a Site Group, all alarms in a Site, or all alarms on a Server.

An Alarm Summary provides summary information such as the number of alarms or communication errors that exist for all managed servers that are underneath the selected node of the Installations tree.

From any level of the Installations tree, you can easily and quickly advance (that is, drill down) to the specific alarm of interest and take any necessary corrective action. To configure alarms, refer to "Configure Alarms" on page 29.

You are notified by email when an alarm exists if you are added to an alarm notification profile that is assigned to the alarm.

Information is presented in the following topics:

- View a Site Group Alarm Summary, page 9
- View a Site Alarm Summary, page 12
- View a Server Alarm Summary, page 16

View a Site Group Alarm Summary

You can view a summary of the active alarms and communication errors at each Site or Site Group within a Site Group. (You can have a Site Group defined beneath another Site Group.) Use the Alarms Status page to view the alarm summary information for a Site Group.

NOTE The Enterprise node is a special instance of a Site Group. Click the Enterprise node to view alarm summaries for all Site Groups or Sites in your enterprise.

When you view a Site Group Alarm Summary, the right hand pane displays one entry (row) for each Site Group and Site that exists beneath the selected Site Group node.

Each of these entries contains cumulative information for the managed servers at the Site Group or Site node. If you want to see alarm summary information for individual managed servers, you must select a Site or Server node.

To view an alarm summary for a Site Group:

1. Click **System Management > System Monitor > Alarm Status**. The alarm status window displays.

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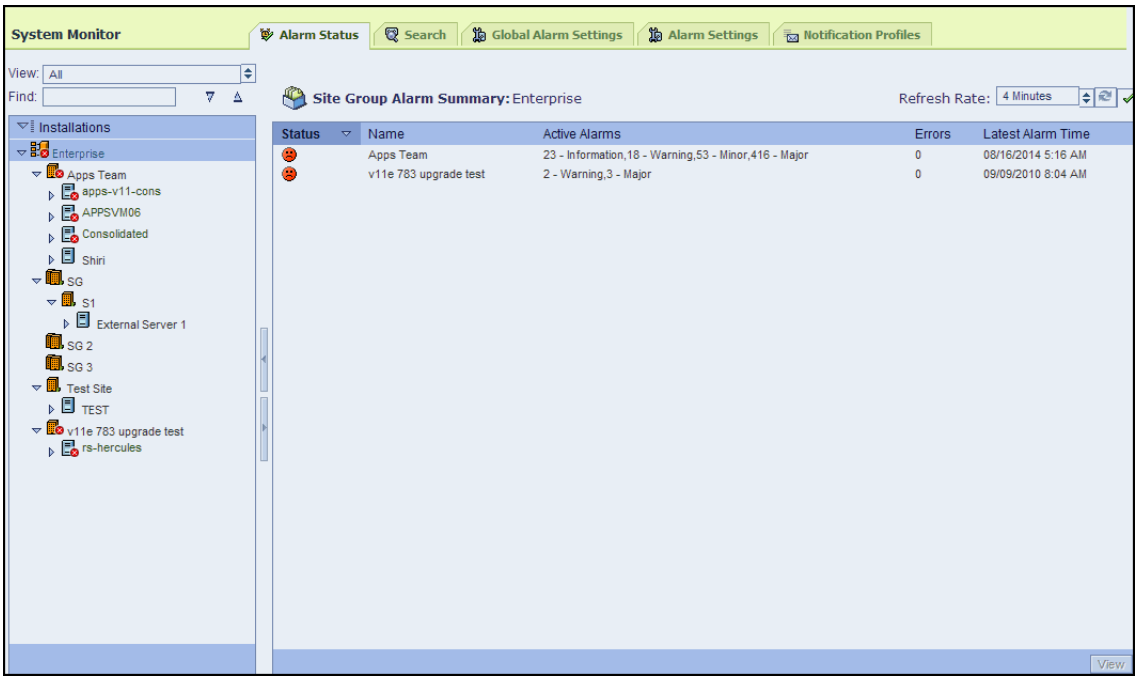
2 In the Installations tree (left pane), select any Site Group (including the Enterprise node). A summary of the alarms that have been triggered at each Site Group or Site within the Site Group displays in the right pane.



3 Review information in the Site Group Alarm Summary according to the following:

NOTE After you resolve and acknowledge an alarm, the alarm may continue to appear as an active alarm in the Enterprise Manager user interface for up to one minute, even though the alarm condition no longer exists. It takes approximately one minute for the Enterprise Manager to detect that the alarm has changed on the managed server.

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Field	Description
Installations (left pane)	Lists the Enterprise, Site Groups, and Sites. Click the triangle beside each name to expand or collapse the Enterprise, Site Group or Site. Click any item to view its details in the right pane.
Status	Shows the status of alarms by color: either green (happy face), red (unhappy face), or grey (expressionless face). The color of the status icon indicates whether the number of active alarms on managed servers under the selected Installation tree node is either below or above the error threshold counts specified for each alarm priority in the **Global Alarm Settings** tab. • **Green icon** - Indicates the number of active alarms is below the error threshold counts on each individual server within the selected Installations tree node. • **Red icon** - Indicates the number of active alarms of any one alarm priority is equal to or above the specified error threshold counts on one (or more) server(s) within the selected Installations tree node. This icon also displays if alarms of the severity level “Unknown” appear in the **Active Alarms** column. See **Active Alarms** below for more information. • **Grey icon** - Indicates the Enterprise Manager can communicate with the managed server but cannot receive information from the alarm service. Check the logs on the managed server to determine the problem. You specify error threshold counts in the **Global Alarm Settings** tab. You specify a separate error threshold count for each alarm priority level (Major, Minor, Warning, Information and any user-defined priorities). If the number of active alarms of any one priority exceeds its specified threshold on any one server, the status icon turns red. For example, assume you specify an error threshold count of three for alarms of the Minor priority. If two Minor priority alarms are triggered on a server, the status icon displays green. When the third Minor priority alarm is triggered on that same server, the status icon displays red for the server. The status icon also displays red for any Site Group or Site to which the server belongs. Next, assume that you specify an error threshold count of three for alarms of the Minor priority and you have two servers in a Site. If server 1 has two Minor alarms triggered and server 2 has one Minor alarm triggered, the status icon for each server displays green. The status icon for the Site to which these two servers belong also displays green, because the error threshold count does not equal or exceed three on any one server in the Site. For information on configuring error threshold counts, see Configure Global Alarm Settings and Priorities on page 29.
Name	Shows the name of a Site Group, Site, or Server for which an alarm summary is available.

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Field	Description
Active Alarms	Shows the total number of active alarms of each priority on the managed servers in the Enterprise, or a selected Site Group or Site. The term **Unknown** appears in this column if there are active alarms on a server and the server is from a server release earlier than V11. Server releases earlier than V11 cannot distinguish the priorities of active alarms when reporting alarms to Enterprise Manager and report alarms of all priorities as Unknown. If there are one or more Unknown alarms, the red icon appears under the Status column.
Errors	Shows the number of alarm collection errors. An error occurs when Enterprise Manager cannot collect (retrieve) alarm information from a managed server. If an error exists, the alarm information displayed for the server may not be accurate. You must troubleshoot and resolve this problem. The message associated with the error should provide information to help you troubleshoot the problem. **Note:** An error often indicates a communication problem exists between the Enterprise Manager and the managed server. For more information about communication between Enterprise Manager and managed servers, see "Communication Over HTTP(S)" on page 356.
Latest Alarm Time	Shows the time at which the latest active alarm was triggered from any of the managed servers in the selected Site Group or Site. If there are no active alarms, this value is empty.
Refresh Rate	Shows the rate at which the user's browser refreshes the current page to update the view of the alarm data displayed by Enterprise Manager. Click the Up/Down arrows to choose a refresh rate from the menu (maximum 20 minutes). The default setting is 1 minute. The higher the refresh rate, the better the system performance resulting from less bandwidth and less processor usage. The default setting provides a balance of good alarm reporting and good performance. **Note:** This setting does not control the interval at which the alarm data is gathered from the managed servers. The alarm data is gathered from the managed servers on an interval that is configured in the EM Alarm Collection settings, as noted in "Configure EM Alarm Collection Settings" on page 396.

4 Click any row and then click **View**, or double-click the row, to view information on the next level. The alarm summary for the next level Site Group or Site displays.

View a Site Alarm Summary

You can view a list of every managed server in a Site that has an active alarm or communications error. For example, a Site within the Midwest Site Group might be named Iowa. Clicking the Iowa Site node lists every managed server in the Iowa Site that has an active alarm or communications error.

When you view a Site Alarm Summary, the right hand pane displays one entry (row) for each managed server (with at least one active alarm) that exists beneath the selected

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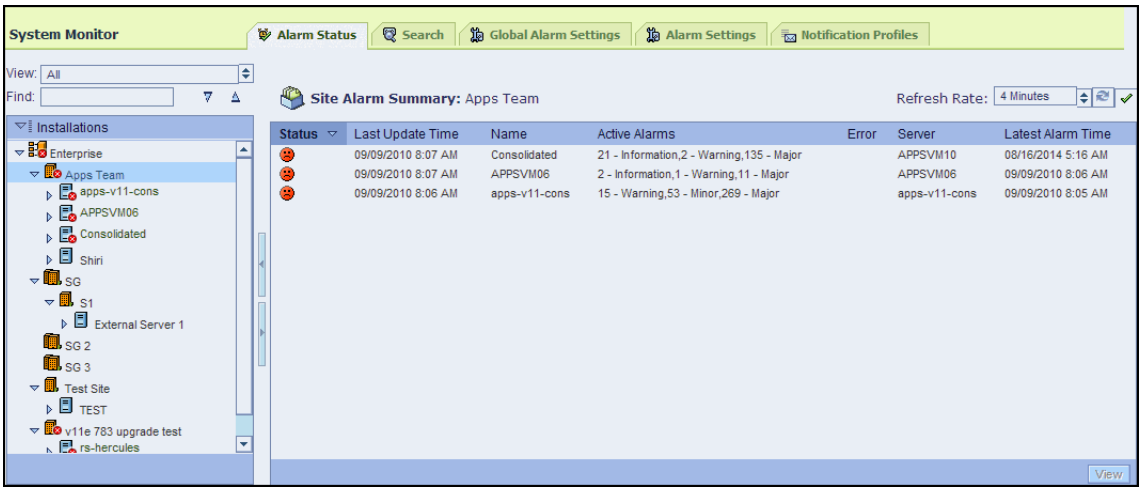
Site node. Even if multiple alarms are triggered on a server, only one entry displays for the server in the Site Alarm Summary.

Each entry contains cumulative information concerning the alarms that are active on one managed server. If the server is a V11 server, the number of alarms of each priority that are triggered on the server appears in the Active Alarms column. If you want to view information about the individual alarms, you must select the server entry and click the **View** button or double-click on the server entry.

Use the Site Alarm Summary window to view the alarm summary information for each managed server in a Site.

To view an alarm summary for a Site:

1. Click **System Management > System Monitor > Alarm Status**.
2. In the Installations tree (left pane), select a Site. A list of all managed servers in that Site that have an alarm(s) or communications error triggered appears in the right pane.



3. Review information in the Site Alarm Summary according to the following:

NOTE

After you resolve and acknowledge an alarm, the alarm may continue to appear as an active alarm in the Enterprise Manager user interface for up to one minute, even though the alarm condition no longer exists. It takes approximately one minute for the Enterprise Manager to detect that the alarm has changed on the managed server.

Field	Description
Installations (left pane)	Lists Site Groups, Sites and Servers. Click the triangle beside each name to expand or collapse the Site Group, Site, or Server. Click any item to view its details in the right pane.

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Field	Description
Status	Shows the status of alarms by color: either green (happy face), red (unhappy face), or grey (expressionless face). The color of the status icon indicates whether the number of active alarms on managed servers within the selected Installation tree node is either below or above the error threshold counts specified for each alarm priority in the **Global Alarm Settings** tab. • **Green icon** - Indicates the number of active alarms is below the error threshold counts on each individual server within the selected Installations tree node. • **Red icon** - Indicates the number of active alarms of any one alarm priority is equal to or above the specified error threshold counts on one (or more) server(s) within the selected Installations tree node. This icon also displays if alarms of the severity level "Unknown" appear in the **Active Alarms** column. See **Active Alarms** below for more information. • **Grey icon** - Indicates the Enterprise Manager can communicate with the managed server but cannot receive information from the alarm service. Check the logs on the managed server to determine the problem. You specify error threshold counts in the **Global Alarm Settings** tab. You specify a separate error threshold count for each alarm priority level (Major, Minor, Warning, Information and any user-defined priorities). If the number of active alarms of any one priority exceeds its specified threshold on any one server, the status icon turns red. For example, assume you specify an error threshold count of three for alarms of the Minor priority. If two Minor priority alarms are triggered on a server, the status icon displays green. When the third Minor priority alarm is triggered on that same server, the status icon displays red for the server. The status icon also displays red and for any Site Group or Site to which the server belongs. Next, assume that you specify an error threshold count of three for alarms of the Minor priority and you have two servers in a Site. If server 1 has two Minor alarms triggered and server 2 has one Minor alarm triggered, the status icon for each server displays green. The status icon for the Site to which these two servers belong also displays green, because the error threshold count does not equal or exceed three on any one server in the Site. For information on configuring error threshold counts, see Configure Global Alarm Settings and Priorities on page 29.
Last Update Time	Shows the last time that the alarm summary information was updated. By default, the Enterprise Manager polls the managed servers once every 30 seconds for new alarm summary information. This field shows the Last Update Time that was valid at the moment the screen was refreshed according to the Refresh Rate setting. For information about configuring the interval at which Enterprise Manager polls managed servers for alarm summary information, see "Configure EM Alarm Collection Settings" on page 396.

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Field	Description
Name	Shows the logical name of the managed server on which the alarms or communications errors exist. The name displayed here is specified in the **Name** field of the **System Management > Installations > Settings > Server** window.
Active Alarms	Shows the total number of active alarms of each priority on each managed server in the selected Site. The term **Unknown** appears in this column if there are active alarms on a server and the server is from a server release earlier than V11. Server releases earlier than V11 cannot distinguish the priorities of active alarms when reporting alarms to Enterprise Manager and report alarms of all priorities as Unknown. If there are one or more Unknown alarms, the red icon appears under the Status column.
Errors	Shows the number of alarm collection errors. An error occurs when Enterprise Manager cannot collect (retrieve) alarm information from a managed server. If an error exists, the alarm information displayed for the server may not be accurate. You must troubleshoot and resolve this problem. The message associated with the error should provide information to help you troubleshoot the problem. **Note:** An error often indicates a communication problem exists between the Enterprise Manager and the managed server. For more information about communication between Enterprise Manager and managed servers, see "Communication Over HTTP(S)" on page 356.
Server name	Shows the Server Name of the managed server on which the alarms or communications errors exist. The name displayed here is specified in the **Server Name** field of the **System Management > Enterprise Management > Settings > Server** window.
Latest Alarm Time	Shows the time at which the latest active alarm was triggered on a managed server. If there are no active alarms, this value will be empty.
Refresh Rate	Shows the rate at which the user's browser refreshes the current page to update the view of the alarm data displayed by Enterprise Manager. Click the Up/Down arrows to choose a refresh rate from the menu (maximum 20 minutes). The default setting is 1 minute. The higher the refresh rate, the better the system performance resulting from less bandwidth and less processor usage. The default setting provides a balance of good alarm reporting and good performance. **Note:** This setting does not control the interval at which the alarm data is gathered from the managed servers. The alarm data is gathered from the managed servers on an interval that is configured in the EM Alarm Collection settings, as noted in "Configure EM Alarm Collection Settings" on page 396.

4 Select any row and then click **View** to view the relevant information on that managed server (that is, drill down) or double-click the row. The System Monitor of the selected managed server displays as described in View a List of Active Alarms for a Server or a Server Role on page 17.

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View a Server Alarm Summary

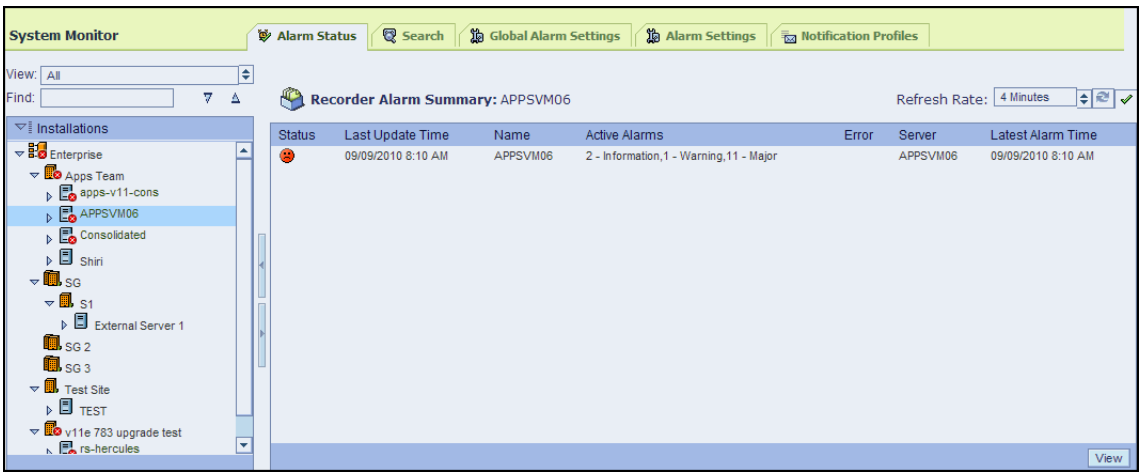
You can view a summary of the active alarms or communications errors on a specific managed server. Use the Server Alarms Summary window to view the alarm summary information for a specific managed server.

When you view a Server Alarm Summary, the right hand pane displays one entry for the managed server. Even if multiple alarms are triggered on a server, only one entry displays for the server in the Site Alarm Summary.

This entry contains cumulative information concerning the alarms that are active on the server. If you want to view information about the individual alarms, you must select the server and click the **View** button or double-click on the server entry.

To view an alarm summary for a managed server:

1. Click **System Management > System Monitor > Alarm Status**.
2. In the Installations tree (left pane), select a managed server. Information on the selected server displays in the right pane.



3. Review the field descriptions according to View a Site Alarm Summary on page 12.
4. Select the managed server and then click **View** to view the relevant information on the server (that is, drill down) or double-click the server.

   When you select View, or double-click the server, the Enterprise Manager Agent (EMA) application or the Recorder Manager (RM) application on the managed server launches and displays a list of alarms on the managed server as described in View a List of Active Alarms for a Server or a Server Role on page 17.

View Active Alarms and Acknowledge Alarms

An active alarm indicates that some condition exists in the system that requires attention.

To view information about specific active alarms, you view a list of all active alarms on a managed server, or on a specific server role. From these lists, you can click on a specific active alarm to read detailed information about the alarm and display the corrective actions for the alarm.

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An active alarm must be acknowledged to make the alarm inactive and move it into alarm history.

Some alarms are acknowledged automatically by the system while other alarms must be manually acknowledged by an administrator. To manually acknowledge an alarm, an administrator first performs the corrective action to address the alarm, and then acknowledges the alarm from the user interface.

You can manually acknowledge alarms one at a time, or you can manually acknowledge multiple alarms simultaneously.

Information is presented in the following topics:

- View a List of Active Alarms for a Server or a Server Role, page 17
- View Active Alarm Corrective Actions and Acknowledge an Alarm, page 20
- Acknowledge Multiple Alarms Simultaneously, page 22
- View or Clear Alarms in Alarm History, page 23

View a List of Active Alarms for a Server or a Server Role

You can view a list of all active alarms on a managed server or a list of all active alarms associated with a specific server role.

These lists of alarms provide preliminary information about each alarm. To view more detailed information about the alarm, select the alarm and click the **View** button or double-click on the alarm. For more information, see View Active Alarm Corrective Actions and Acknowledge an Alarm on page 20.

You must have View Alarms privileges to view the list of alarms on a managed server or to view the list of alarms associated with a specific server role.

To view a list of active alarms for a managed server or a server role

1. Click **System Management > System Monitor > Alarm Status**.
2. In the Installations tree (left pane), select the specific Server node for which you want to view alarms.
3. In the right-hand pane, either click on the row for the managed server and then click the **View Alarms** button, or double-click on the row for the server.

   Performing this step launches the Enterprise Manager Agent (EMA) application or the Recorder Manager (RM) application on the managed server. The EMA or RM

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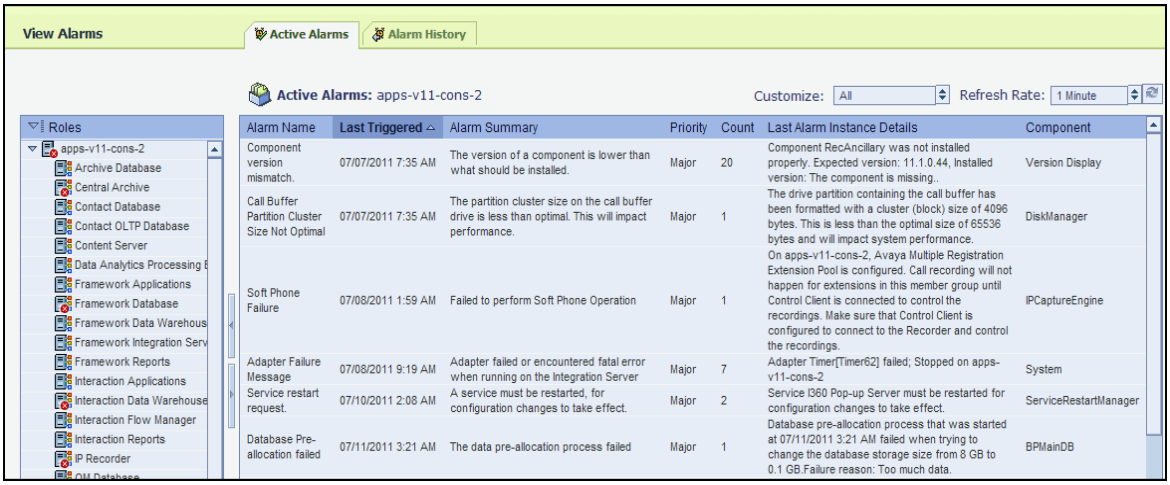
provides information about the individual alarms (both active and inactive) on a managed server.



4 Expand the Server node (click on the small triangle beside the Server) in the left-hand pane of the EMA or RM to view the server roles. There are several user interface options available to help you interpret the alarm information. Notice that:

- if an alarms (red) icon displays on a Server or Server Role node, this indicates that alarms exist.
- if you move and hold your mouse over a Server or Server Role node with an alarms (red) icon, the number of alarms on that server or server role displays like a tool tip.
- an alarms icon on the Server node approximates the total number of active alarms for all server roles on that server. Note that multiple server roles can raise different instances of the same alarm.
- an alarms icon on a Server Role node shows the number of active alarms on that role only.

5 Do one of the following:

- To view a list of all active alarms on a managed server, select the Server node in the left-hand pane. Alarms associated with all roles on the managed server display in the Active Alarms window (right-hand pane).
- To view a list of all active alarms for a specific server role, select a Server Role node in the left-hand pane. Alarms associated only with that server role display in the Active Alarms window.

6 Review information according to the following table.

NOTE If all of the information discussed below does not appear on the screen, the screen was customized to exclude the missing information. Use the **Customize** option above the screen to display the missing information.

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Item	Description
Alarm Name	The name of the alarm.
Last Triggered	The date and when the alarm was last raised (triggered). For multiple instances of an alarm, the latest trigger time of the alarm instance is displayed.
Alarm Summary	A summary of the condition causing the alarm.
Priority	The priority specified for the alarm. Alarm priorities are specified for individual alarms from the **System Management > System Monitor > Alarm Settings** tab. For more information, see "Edit an Alarm" on page 37.
Count	Instance count. Number of occurrences/instances of the alarm that are active.
Last Alarm Instance Details	Summary description of the alarm. This is the common description for all instances and not based on a specific instance. For a detailed, instance-specific description, select the alarm and then click **View** the alarm, or review the alarm in alarm history.
Component	The system component that triggered the alarm. Some alarms can be triggered by more than one component.
Refresh Rate	Shows the current setting for refresh rate in minutes. Choose 1,2,3,4,5,10,15, or 20 minutes as the automatic refresh rate. At these times, the EMA application updates the status of the alarms on the managed server. Default is 1. You can also click the **Refresh** icon (inverted arrows beside the selection box) to update alarms immediately, without waiting for the automatic refresh cycle.

7 Do one of the following:

- Select an alarm and click **View** to view additional details of the alarm.
- Perform the corrective action for the alarm. Then select the alarm and click **Acknowledge** to acknowledge the alarm.

  Note that you can acknowledge multiple alarms simultaneously. See "Acknowledge Multiple Alarms Simultaneously" on page 22 for detailed information about acknowledging alarms.

Launching Multiple Recorder Manager Applications

If you want to view the lists of active alarms on several different managed servers, you may want to open the EMAs or RMs for several different servers and have all of the EMAs or RMs open at the same time. By default, each EMA or RM will open in a separate window on your computer. You can configure your Internet Explorer browser so that each instance of the EMA or RM automatically opens in its own separate tab in Internet Explorer. This configuration helps you more efficiently manage and navigate between all of the EMAs or RMs when several instances are open simultaneously. To configure Internet Explorer so that each instance of EMA or RM opens in a separate tab, follow this procedure:

1 Open Internet Explorer.

2 Select the **Tools > Internet Options** menu.

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3 Select the **General** tab.

4 In the **Tabs** section of the General tab, select the **Settings** button.

5 In the **When a pop-up is encountered** section, select the **Always open pop-ups in a new tab** option.

6 Each Recorder Manager application will now open in a separate tab in Enterprise Manager.

View Active Alarm Corrective Actions and Acknowledge an Alarm

View the details of an active alarm to obtain the most complete information about the alarm, including the corrective action for the alarm.

Before you can acknowledge an alarm, you must perform the corrective action that corrects the condition causing the alarm.

To view alarm details and corrective actions:

1 Click **System Management > System Monitor > Alarm Status**.

2 In the Installations tree (left pane), select the Server node for which you want to acknowledge alarms.

3 In the right-hand pane, either click on the row for the managed server and then click the **View Alarms** button, or double-click on the row for the server.

Performing this step launches the Enterprise Manager Agent (EMA) or Recorder Manager (RM) application on the managed server. The EMA or RM provides information about the individual alarms (both active and inactive) on a managed server.

4 In the Roles (left-hand) pane, select a Server or Server Role node to view the list of alarms for the selected server or server role.

5 In the right-hand pane, select an alarm and then click **View**, or double-click the alarm.

NOTE The **View** button is unavailable if more than one alarm is selected.

If you selected the Server node, the alarm details window displays all instances of the alarm. If you selected the Server Role node, the alarm details window displays the single instance of the alarm that applies to that server role.

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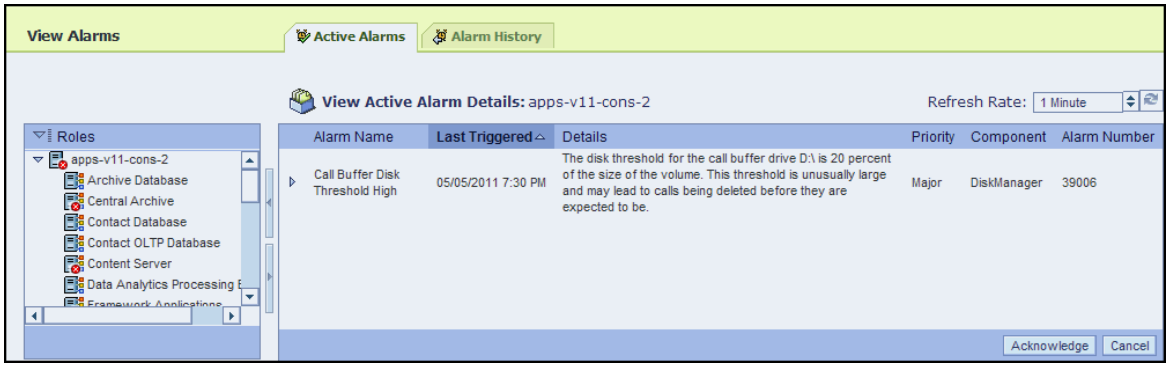

6 Review fields according to the following:

Item	Description
Corrective Action Steps	To view a drop-down list containing the corrective action steps for the alarm, click on the small right-pointing arrow that appears just to the left of the alarm name. Follow these steps to correct the alarm condition. Once the alarm has been corrected, you can acknowledge the alarm to place it in Alarm History.
Alarm Name	The name of the alarm.
Last Triggered	The date and time when the alarm was last raised (triggered). If there are multiple instances of an alarm, the latest trigger date and time of the alarm instance is displayed.
Details	The specific description of the alarm. This is the most detailed information available about this specific instance of the alarm.
Priority	The default priority/severity of the alarm. To change the priority level of an alarm, see Edit an Alarm on page 37.
Component	The component on the selected managed server or server role that triggered the alarm.
Alarm Number	The unique numerical identifier of the alarm.
Refresh Rate	Shows the current setting for refresh rate in minutes. Choose 1,2,3,4,5,10,15, or 20 minutes as the automatic refresh rate. At these times, the EMA or RM application updates the status of the alarms on the managed server. Default is 1. You can also click the **Refresh** icon (inverted arrows beside the selection box) to update alarms immediately, without waiting for the automatic refresh cycle.

7 Do one of the following:

- Perform the corrective action for the alarm, and then click **Acknowledge** to make the alarm inactive and transfer it to alarm history.

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Note the following:

- If an alarm affects multiple server roles, acknowledging the alarm from one server role acknowledges it from all server roles.
- If there are multiple instances of the same alarm, acknowledging one instance of the alarm acknowledges all instances of the alarm.
- You can only acknowledge one alarm from this screen. You can simultaneously acknowledge multiple alarms from the Active Alarms tab, as noted in "Acknowledge Multiple Alarms Simultaneously" on page 22.

- Click **Cancel** to return to the list of alarms without making any changes.

Acknowledge Multiple Alarms Simultaneously

You can acknowledge multiple alarms, or all alarms, simultaneously.

Under normal circumstances, you should acknowledge alarms one-at-a-time using the procedure described in "View Active Alarm Corrective Actions and Acknowledge an Alarm" on page 20. This procedure provides a convenient way to clear alarms if there are too many alarms to acknowledge one-at-a-time.

The Acknowledge Alarms privilege is required to acknowledge alarms.

To simultaneously acknowledge multiple, or all, alarms:

1. Click **System Management > System Monitor > Alarm Status**.
2. In the Installations tree (left pane), select the Server node for which you want to view alarms.
3. In the right-hand pane, either click on the row for the managed server and then click the **View Alarms** button, or double-click on the row for the server.

   Performing this step launches the Enterprise Manager Agent (EMA) or Recorder Manager (RM) application on the managed server. The EMA or RM provides information about the individual alarms (both active and inactive) on a managed server.
4. In the Roles (left-hand) pane, select the Server or Server Role node for which you want to acknowledge multiple (or all) alarms.
5. Do one of the following:
   - To simultaneously acknowledge multiple (but not all) alarms, hold down the **Ctrl** key, select each alarm you want to acknowledge, and click **Acknowledge**.

     You can also select a range of alarms by holding down the **Shift** key, and clicking on the top and bottom alarm in the range.
   - To simultaneously acknowledge all active alarms, click **Select All**, then click **Acknowledge**.

     If you decide not to acknowledge all alarms, click **Select None** to de-select all alarms.

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6 You are prompted to verify that you have started all services indicated by the alarms you are acknowledging. (Many alarms require you to restart a specific service before acknowledging the alarm.)

Perform this verification, and then click **OK**.

NOTE If you acknowledge multiple alarms simultaneously, it may take time for the system to process the acknowledgements. In this case, a message appears at the top of the screen indicating the system is processing the alarms. When the acknowledgement processing completes, the alarms will move from the Active Alarms tab to the Alarm History tab.

View or Clear Alarms in Alarm History

View alarm history to review alarms that have been corrected and acknowledged, and now are in an inactive state. From this window, you can view inactive alarms for each managed server or server role, and clear all displayed alarms. You cannot clear only alarms that you select; you must clear all alarms in alarm history or clear none. A maximum of 1000 inactive alarms are retained in alarm history.

You need Edit Alarms privilege to clear alarms from alarm history.

To view or clear alarms in alarm history:

1 Click **System Management > System Monitor > Alarm Status**.

2 In the Installations (left-hand) pane, select the Server node for which you want to view the alarm history.

3 In the right-hand pane, either click on the row for the managed server and then click the **View** button, or double-click on the row for the server.

Performing this step launches the Enterprise Manager Agent (EMA) application or the Recorder Manager (RM) application on the managed server. The EMA or RM provides information about the individual alarms (both active and inactive) on a managed server.

4 Click the **Alarms History** tab. All inactive alarms for the selected managed server or server role display.

The same alarm history displays regardless of whether you have selected the Server node or a Server Role node in the right-hand pane. That is, Alarm History is constant and does not vary according to the selected server or server role.

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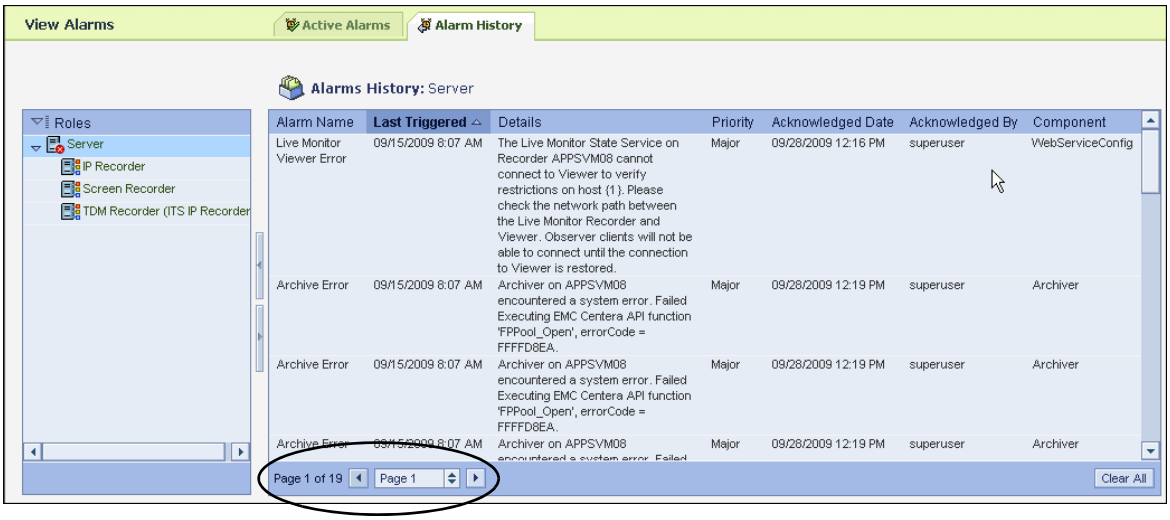

5 Review fields according to the following:

Item	Description
Alarm Name	The name of the alarm.
Last Triggered	The last date and time the alarm was raised (triggered). For multiple instances of an alarm, the latest trigger date and time of the alarm instance is displayed.
Details	Displays detailed information about the specific instance of the alarm.
Priority	The priority specified for the alarm. Alarm priorities are specified for individual alarms from the **System Management > System Monitor > Alarm Settings** tab. For more information, see "Edit an Alarm" on page 37.
Acknowledged Date	The date and time when the alarm was placed into alarm history (that is, acknowledged).
Acknowledged By	The name of the component (for automatic acknowledgement) or user (if manual acknowledgement) that acknowledged the alarm. **Note:** Some alarms are automatically acknowledged by components.
Component	The component or user on the selected managed server or server role that caused the alarm to be triggered.

6 Do one of the following:

- Use the **Page** navigation controls at the bottom of the right-hand pane, (circled) if present, to view additional pages of inactive alarms (that is, alarms that have already been acknowledged). You can change the number of records that display per page in **Preferences**.
- Click **Clear All** with a managed server or server role selected to delete all alarms for that selection from the alarms history window.

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Search for Servers with Alarms

Use the System Monitor Search window to display a list of selected managed servers within the enterprise. You can use the **Find** option in the Search tab to locate a specific managed server in the displayed list. When you locate a particular managed server, the alarm summary for that server displays in the right-hand pane.

The primary purpose of the Search tab is to provide you with a way to easily monitor the alarm activity on a specific set of managed servers. This tab allows you to create a view (or filter) that displays only specific servers in the environment.

For example, assume you have an environment consisting of 60 managed servers that are installed at six different Sites. A situation arises where you need to monitor the alarm activity on server1, server8, server28, server32, and server56.

You can use the Search tab to create a search filter that displays only those five servers (with no Site Group or Site nodes displayed). When you want to monitor those servers, you go to the Search tab, select the search filter to quickly view the alarm summaries for each of those five servers (and does not display any Site Group or Site nodes). This capability prevents you from having to locate each server individually in the Installations tree and view the servers one-at-a-time.

NOTE Search filters are created on a per-user basis. For example, if User A logs in and creates an alarm search filter, the search filter is only visible to User A. If User B logs in, User B cannot see or use the search filter created by User A.

To use the Search tab features, see these topics:

- Search Using an Existing Search Filter, page 25
- Create, Edit, or Delete a Search Filter, page 27

Search Using an Existing Search Filter

The **All** search filter exists by default. The **All** search filter lists every managed server installed in the enterprise. You can enter text in the **Find** window to search for a specific server within the list of all servers. You can also create other, more specific, search filters as discussed in Create, Edit, or Delete a Search Filter on page 27.

You can search for servers using the **All** search filter or any other search filter you have previously created.

To search for a server using an existing search filter:

1. Click **System Management > System Monitor > Search.** The Servers (left) pane is replaced with a Search window populated with results from the previous search.

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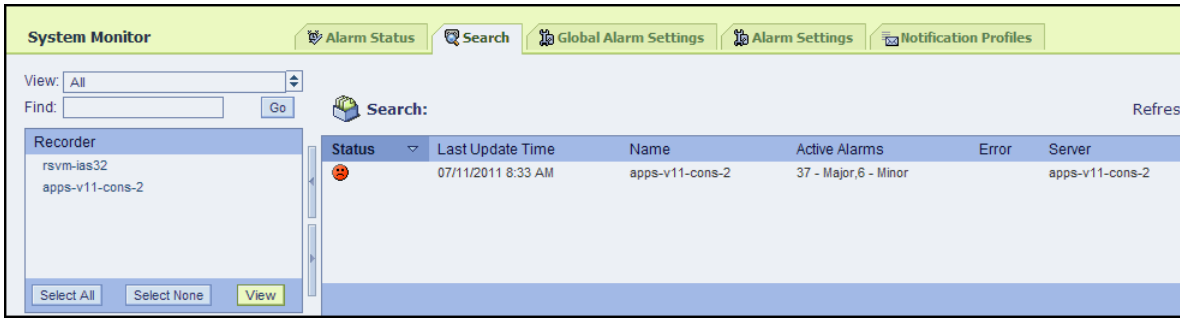
If no filter was chosen previously, the **All** filter is selected by default and a list of all managed servers in the Enterprise displays.



2 To search, do the following in the left pane:

Note: Only managed servers to which you have access privileges display.

- In the **View** window, choose one of the existing search filters.
- With the appropriate view selected, type search criteria in the **Find** area, such as the name or a few letters in the server name, and then click **Go**.

  If you enter only a few letters or numbers of the server name, the search feature will find any server that includes those letters or numbers anywhere in its name. For example, typing only the number 10 in the Find area would find both a server named “server_10” and a server named “10_server”.

3 Do one of the following:

- In the right-hand pane, select a single server and then click the **View** button at the bottom of the right-hand pane. The list of active alarms for the selected server launches and you can work with the alarms as described in View Active Alarm Corrective Actions and Acknowledge an Alarm on page 20.
- Click the **Select All** button at the bottom of the left-hand pane to select all servers, including servers that were not returned by the search, and then click the **View** button at the bottom of the left-hand pane.

  The right-hand pane displays the alarm summary information for all of the selected servers. You can work with the alarm summaries as discussed in “View a Server Alarm Summary” on page 16.
- Click **Select None** at the bottom of the left-hand pane to deselect all of the servers returned by the search.

4 Click the **View** button at the bottom of the left-hand pane. A list of alarm summaries displays according to your selection in the previous step.

5 Once you have located a server, you can select the server and click the click View button at the bottom of the right-hand pane (or double-click the server) to work with the server, as described in View Active Alarm Corrective Actions and Acknowledge an Alarm on page 20.

If the currently selected filter has the **Alarms and Errors Only** option selected as described in Create, Edit, or Delete a Search Filter on page 27, the right pane displays only those servers with alarms or errors.

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Create, Edit, or Delete a Search Filter

You can create, edit, or delete alarm search filters.

Use the **Create Filter** option in the View drop-down box to create a custom search filter.

Normally, you create a custom filter so that you can quickly display a list of specific servers in the Search tab to monitor the alarm activity on those servers.

When you select the **Create Filter** option, the **Create Filter** window displays the Installations tree hierarchy and the alarm search options.

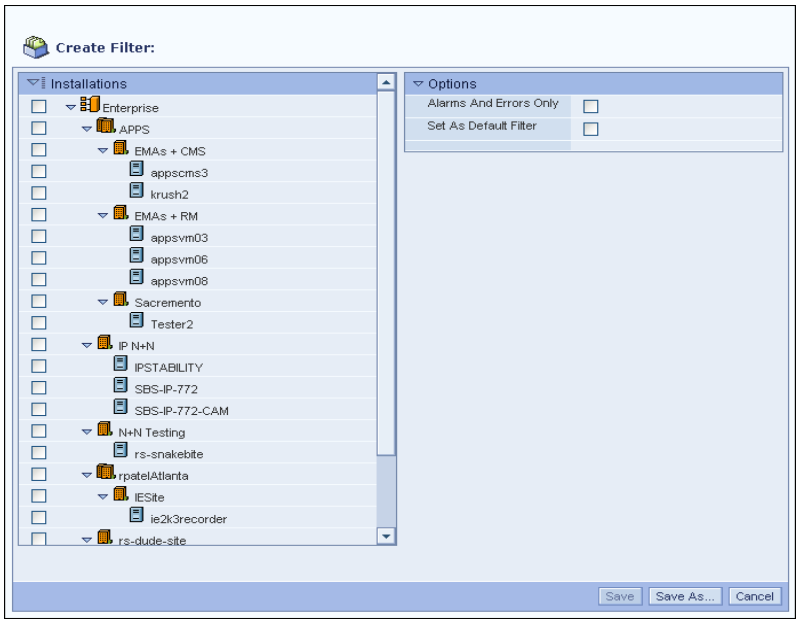
The Site Groups, Sites, or specific Servers that you select from the Installations tree are the ones that will display in the Search tab when you select the search filter.

Use the alarm search options to determine whether the search returns only servers that have alarms and errors, or to set the search filter as the default search filter.

Use the **Edit Filter** option in the View drop-down box to edit a search filter. Edit the filter by adding or deleting Site Groups, Sites, and Servers from the search criteria of an existing alarm search filter, or by changing the search options.

To create a new search filter:

1. Click **System Management > System Monitor > Search.**
2. Click the arrows beside **View** and choose **Create Filter**.



3. In the Installations tree (left pane), select the Site Groups, Sites, or individual Server nodes from the Installations tree. The items you select are the items that display when this filter is selected.

   You can use the **Find** field in the Search tab to locate specific servers within the displayed list.

4. Under **Options**, you can select one, both, or neither of the following:
   - **Alarms and Errors Only** - Select this option if you want the filter to display only those managed servers with alarms triggered or with communications

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errors. Deselect this option if you want the filter to display both servers with alarms triggered and servers without alarms triggered.

- **Set as Default Filter** - Select this option to have a filter act as the default filter. Setting this option makes that filter the filter that is automatically selected when you first access the **System Management > System Monitor > Search** tab.

5 Click **Save As**. The application adds the new filter name to the filters in the View list.

To edit or delete a search filter:

1 Click **System Management > System Monitor > Search.**

2 Click the arrows beside **View** above the left pane, and select the search filter you want to edit.

3 Click the arrows beside **View** above the left pane, and then choose **Edit Filter**. The Edit Alarms Filter window displays showing the Site Groups, Sites, and Servers and the alarm options that are included in the current search filter.

4 In the Installations tree (left pane), select different Site Groups and Sites to filter by clicking the checkbox beside each Site Group, Site or Server node. The nodes you select are the nodes that are included in any search conducted using this search filter.

5 Under **Options**, select one, both, or neither of the following:

- **Alarms and Errors Only** - Select this option if you want the search filter to display only those managed servers with alarms triggered or with communications errors. Deselect this option if you want the search filter to return all servers that match the search criteria entered in the Find field, regardless of whether the server has alarms triggered.
- **Set as Default Filter -** Select this option to have a filter act as the default filter. Setting this option makes that filter the filter that is automatically selected when you first access the Alarms Search page.

6 Do one of the following:

- Click **Save** to save an edited version of the search filter. The application updates the filter and the list of servers in the left pane is updated based on the criteria you selected in steps 4 and 5.
- Click **Save as** to save the edited version of the search filter as a new search filter with a new name. The contents of the left pane change to reflect the criteria you selected in steps 4 and 5, and the new filter name is added to the filters in the View list.
- Click **Delete** to delete the selected search filter. The filter is deleted and the contents of the left pane change to display all the managed servers to which you have View Alarms privileges.

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Configure Alarms

Configuring alarms involves configuring global alarm settings and editing individual alarm settings. Global alarm settings apply by default to all alarms. After configuring the global alarm settings, you can edit individual alarm settings to further define a specific alarm's behavior.

The global alarm settings include basic alarm settings, such as whether an audible alarm sounds and whether an email notification is sent when any alarm is triggered. Global alarm settings also include alarm priority level definitions and the SNMP notification settings that apply to all alarms.

You can edit individual alarms to further configure the behavior of a specific alarm. For example, you can specify whether an audible alarm sounds for an alarm and assign a priority level to an alarm. You can schedule the times of day in which a specific alarm is in effect. You can also assign notification profiles to an alarm to determine who is notified by email, and which SNMP nodes receive SNMP traps, when an alarm is triggered. Editing individual alarms can override some of the global alarm settings.

NOTE Enterprise Manager transmits alarm configuration changes to the managed servers in configuration messages. The Enterprise Manager Agent (EMA) on the managed server implements the alarm configuration changes and allows you to see details about the individual alarms. For more information, see "Enterprise Manager Agent and Alarms" on page 364.

Information is presented in the following topics:

- Configure Global Alarm Settings and Priorities, page 29
- View or Reset Alarms, page 33
- Edit an Alarm, page 37
- Edit Multiple Alarms Simultaneously, page 42
- Create, Edit, and Delete Alarm Filters, page 45

Configure Global Alarm Settings and Priorities

Configure the global alarm settings to specify the settings that apply by default to all alarms.

You can configure all of the following from the global alarm settings:

- Whether an audible notification occurs when any alarm is triggered.
- Whether a notification email is sent for alarms that are triggered. (The alarm must have a notification profile assigned to it for a notification email to be sent.)
- Whether an SNMP notification (trap) is sent to your Network Management System for every alarm that is triggered. You can also specify the SNMP nodes that can be assigned to notification profiles so that they can receive the SNMP traps.
- The maximum number of days that an acknowledged alarm remains in alarm history, unless cleared by a user.
- The alarm priority levels that are available to be assigned to all alarms. You can also specify the error threshold count for each alarm priority level.

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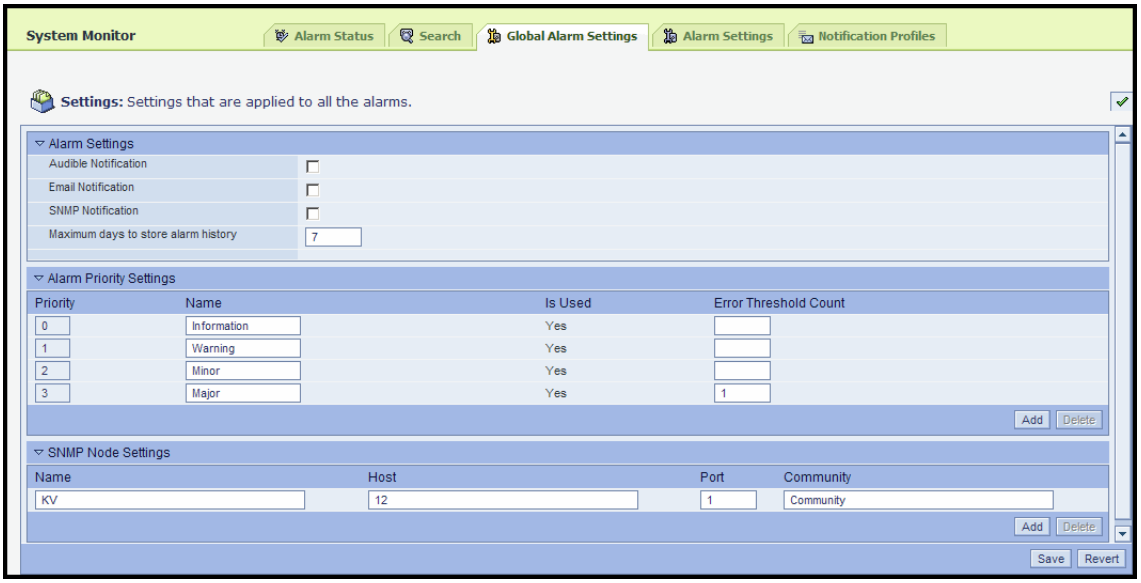
You need Edit Global Alarm Settings privileges to edit global settings.

To configure global alarm settings and priorities:

1 Choose **System Management > System Monitor** > **Global Alarm Settings**.



2 Complete the fields in the **Global Settings** window as follows:

Field	Description
Alarm Settings	
Audible Notification	Select this option if you want a sound to accompany every alarm that is triggered. This field is selected by default. If this option is selected, you can still disable audible notification at an individual alarm level, as described in Edit an Alarm on page 37. If you disable this setting (uncheck it), audible notification for all alarms is disabled and this behavior cannot be changed at the individual alarm level. (The audible notification field is greyed-out at the individual alarm level.)
E-Mail Notification	Select this option if you want email notifications to be sent when an alarm is raised. This field is selected by default. The notification profiles assigned to each alarm determine who receives the e-mail notifications, as described in Edit an Alarm on page 37. This field is not selected by default.
SNMP Notification	Select this option if you want SNMP (Simple Network Messaging Protocol) traps to be sent to the Network Management System each time an alarm is triggered. The notification profiles assigned to each alarm determine the SNMP nodes to which SNMP traps are sent, as described in Edit an Alarm on page 37. This field is not selected by default.

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Field	Description
Max days to store alarm history	Type the maximum number of days that an alarm should be displayed in alarm history. The default for this field is 7 days. After this period, the inactive alarm (in alarm history) is deleted.
Alarm Priority Settings	
Priority	Set the numeric value of the alarm priorities. Generally, a priority of zero applies to alarms that have the least severity or effect on system performance. The priority value increases with the severity of the alarm. The default settings are: 0 - Information 1 - Warning 2 - Minor 3 - Major
Name	Shows the name of the alarm priority. Four default alarm priorities are given: Information (0), Warning (1), Minor (2), and Major (3), representing severity levels for each alarm. You can customize the default alarm priorities at any time by renaming the default fields and values then clicking **Save**. You can also add new alarm priorities as well as delete unassigned alarm priorities that you have created. **To add an alarm priority:** 1. Click **Add**. A new alarm priority with a value of 9 (the highest) is added. 2. Enter a name, such as *Critical* for the new alarm priority. 3. Change the priority value from 9 to the desired value, such as *4*. In this example, you now have 5 alarm priority levels. 4. Click **Save.** You can now apply the new *Critical* priority to alarms, as described in Edit an Alarm on page 37. **To delete an alarm priority that you added:** 1. Place your cursor in the Priority or Name value of an alarm priority setting that you've added. (You cannot delete any of the four default alarm priorities.) 2. Click **Delete.** 3. Click **Save**.
Is used	Shows **Yes** if the alarm priority has been applied to an alarm, and **No** if it has not been applied to an alarm. To apply alarm priorities to alarms, see Edit an Alarm on page 37.

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Field	Description
Error Threshold Count	For each alarm priority, specify the active alarm threshold at which the alarm icon on the **Alarm Status** tab changes from green (happy face) to red (unhappy face indicating a serious problem on a server). For example, if you specify a value of 3 for the Minor alarm priority and 1 for the Major alarm priority, the alarm icon on the **Alarm Status** tab will turn red if either 3 or more Minor alarms, or 1 or more Major alarms are triggered on one managed server. Note that the Error Threshold Count that you specify applies to the number of alarms that exist on a **single** server. For example, assume you specify a value of 3 for the Minor alarm priority. If you have two servers in a Site, and two Minor priority alarms exist on one server and one Minority priority alarm exists on the other server, the alarm icon for that Site will display green. The alarm icon displays green because no single server in the Site has three or more Minor priority alarms. If you leave this setting blank (or specify zero) for any alarm priority, the alarm icon will not turn red no matter how many alarms of that priority are triggered. The default setting is 1 for the Major priority level and blank (or zero) for all other alarm priorities. When configuring this setting, specify the number of alarms of each priority that can acceptably exist on any single server in your enterprise. The intent of this feature is to alert you by displaying a red icon in the **Alarm Status** tab when any single server has a serious problem. Alarm priorities are assigned to individual alarms from the **Alarm Settings** tab. For more information, see Edit an Alarm on page 37. A server communication error is considered to be the same as an active alarm with a Major priority. If this setting specifies 1 for the Major priority and a server communication error occurs, the alarm status icon turns red on the **Alarm Status** page.

SNMP Node Settings.

Add an SNMP node to make it available to be assigned to a notification profile. SNMP nodes assigned to notification profiles receive SNMP traps when alarms are triggered.

To add an SNMP node, click the Add button and complete the fields below.

Name	Type the name of an SNMP trap configuration, such as Atlanta SNMP server.
Host	Type the hostname or IP address of the server to which the SNMP trap message will be sent.

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Field	Description
Port	Type the port number on which the SNMP server listens for SNMP traps. The usual default port for this purpose is port 162. The port you enter here must be the same port on which your SNMP server is configured to listen for trap messages.
Community	Optionally type the name used to identify the group to which SNMP devices and management stations belong. Some Network Management Systems (NMS) ignore traps that do not include community strings. For this kind of NMS, the community name acts as a weak password that is required for the NMS to consider the trap valid. Such an NMS ignores traps that don't have the appropriate community string.

3 Do any of the following:

- In the **Alarm Settings** area, enable any options appropriate for your environment (Audible Notification, Email Notification, SNMP Notification, Maximum days to store alarm history).
- In the **Alarm Priority Settings** area, enter the **Error Threshold Count** values that determine when the alarm icon on the **Alarm Status** tab displays red or green.
- In the **Alarm Priority Settings** area, click **Add** to append a new alarm priority level to the list, using the next available number. The **Is used** column shows **No** until the priority is assigned to an alarm.
- In the **Alarm Priority Settings** area, click **Delete** to delete an alarm priority that you have created. You cannot delete any of the four default alarm priorities.
- In the **SNMP Node Settings** area, click **Add** to append the settings for a new SNMP node, as described above. Click **Delete** to delete one or more selected SNMP Node setting(s).

4 Click **Save**.

View or Reset Alarms

You can view lists of alarms from the Enterprise, Site Group, Site, or Server nodes of the Installations tree.

The node of the Installations tree that you select determines the list of alarms that display in the right-hand pane. For example:

- Select the Enterprise node to view a list of all alarms that are defined on every managed server in the enterprise.
- Select a Site Group node to view a list of all alarms that are defined on every managed server in the selected Site Group.
- Select the Site node to view a list of all alarms that are defined on every managed server in the selected Site.
- Select the Server node to view a list of all alarms that are defined on the selected server.

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The list of alarms provides you with basic information about each of the alarms, such as an alarm's priority, a basic description of the alarm, whether the alarm is enabled, and the server roles to which the alarm applies.

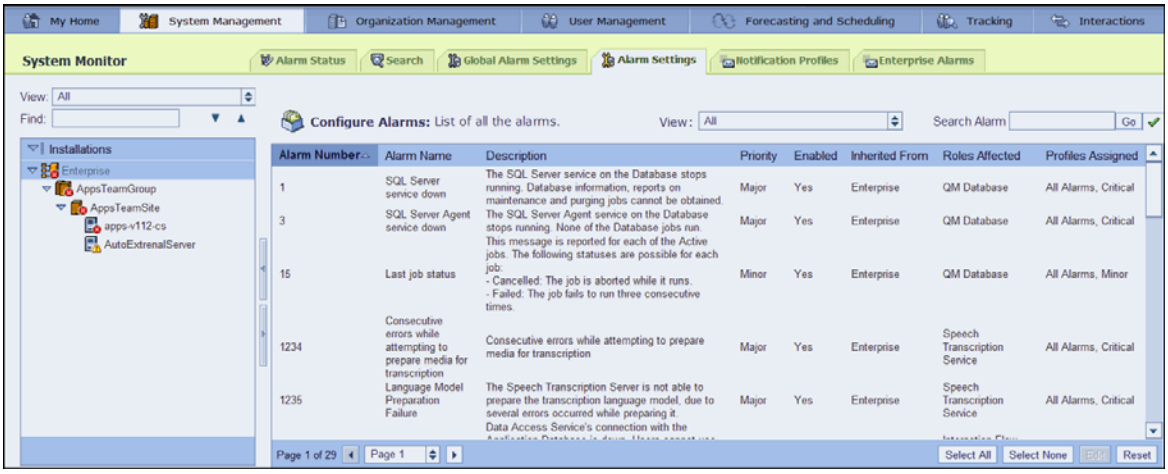
You can also reset all of the alarms. Resetting the alarms restores alarm settings for all managed servers installed in the enterprise to the default alarm settings. Note the following about resetting all alarms:

- You must reset the settings for all alarms. You cannot select individual alarms and reset the settings for only the selected alarms.
- All of the settings on the Configure Alarm screen for every alarm are restored to their default settings. For more information about these settings, see Edit an Alarm on page 37.
- Alarm Notification Profiles that you have created or edited are not deleted or changed. However, the Notification Profiles that are assigned to each alarm (on the Configure Alarms screen) are restored to their default settings.
- In the Global Alarm Settings tab, the settings under the Alarm Settings heading are restored to their default settings. However, the settings under the Alarm Priority Settings heading and the SNMP Node Settings heading are not restored to their default settings. For more information, see Configure Global Alarm Settings and Priorities on page 29.

To view a list of all alarms:

1 Click **System Management** > **System Monitor** > **Alarm Settings**.

The Configure Alarms screen displays a list of the alarms according to the **View** (filter) selected in the upper-right corner of the screen. The **All** filter, which displays all servers and all alarms, is selected by default until you declare another default filter, as described in Create, Edit, and Delete Alarm Filters on page 45.



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2 Select the node of the Installations tree for which you want to view a list of alarms:

- Select the Enterprise node to view a list of all alarms on all managed servers installed in the enterprise.
- Select a Site Group node to view a list of all alarms on all managed servers installed in a Site Group.
- Select a Server node to view a list of all alarms on a specific managed server. (The alarms listed depend on the server roles active on that server.)

3 Review fields (all read-only) according to the following:

Field	Description
View	Use the View drop-down menu above the screen to create, edit, or delete an alarm filter. For more information, see "Create, Edit, and Delete Alarm Filters" on page 45.
Search Alarm	Use the Search Alarm field in the upper-right corner to search for terms or phrases that exist in either the **Alarm Name** or the **Description** fields (described below) of the Configure Alarms screen. The search returns all instances of the entire phrase that is entered in the Search Alarm field. For example, if you enter "playback attempts" in the search field and click **Go**, the search returns every alarm that includes the entire phrase "playback attempts" in either its Alarm Name or Description field. Note that such a search does **not** return alarms that have only the term "playback" or only the term "attempts" in the Alarm Name or Description field. (The search is not a full-text index search and it finds only the instances of the entire phrase that is entered in the search field.) To search for all alarms that include only the term "playback" in the Alarm Name or Description fields, you would enter only the term "playback" in the search field and click **Go**.
Alarm Number	The unique numerical identifier of the alarm. Click in the column heading to sort the alarm numbers in ascending (lowest numbered alarm appears at the top) or descending (highest numbered alarm appears at the top) order.
Alarm Name	The name of the alarm.
Description	A brief description of the alarm.
Priority	The priority/severity level for this alarm. Use the procedure described in Edit an Alarm on page 37 to change the priority of an alarm.
Enabled	Shows **Yes** or **No** to indicate if the alarm is active (enabled) or not active. If **No** is selected, the alarm cannot be triggered. Use the procedure described in Edit an Alarm on page 37 to enable or disable an individual alarm.

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Field	Description
Inherited From	The Site Group, Site, or Server node from which the alarm settings (not the actual alarms) are inherited. You can configure settings for individual alarms at the Enterprise, Site Group, Site, or Server node of the Installations tree. The configuration of the alarm can then be inherited by the lower nodes of the Installations tree. (If you click the Save and Apply to Children button after changing an alarm's configuration, the configuration settings are inherited by the lower nodes of the Installations tree.) You can select a lower node of the Installations tree and change (or edit) the configuration for the alarm at the lower node. This capability allows you to configure the same alarm to behave differently for different Site Groups, Sites, or Servers. When you select a node in the Installations tree on the Alarms Settings tab, the Inherited From column shows the node from which the selected node is inheriting the alarm configuration. If the Inherited From field displays the same node that is selected in the Installations tree, the individual alarm settings are configured at the selected node and are not inherited from a higher node in the Installations tree. For more information, see Edit an Alarm on page 37.
Roles Affected	The server roles affected by the condition that this alarm reports. For example, an archive configuration error would affect the functioning of the Centralized Archiving role on a server.
Profiles Assigned	Shows the notification profiles assigned to the alarm. Note that the same alarm can have different Notification Profiles assigned to it at different nodes of the Installations tree. The same Notification Profile can also be configured to notify different employees (via e-mail) and SNMP nodes at different nodes of the Installations tree. To assign notification profiles to an alarm, see Edit an Alarm on page 37.

4 At the bottom of the screen, use the **Page** arrow controls to navigate through the pages of alarms.

5 To edit a specific alarm, follow the steps in Edit an Alarm on page 37.

To reset all alarms to their default settings:

1 Click **System Management** > **System Monitor** > **Alarm Settings**.

2 In the Installations tree (left pane), select the Enterprise node.

3 Click the **Reset** button. A message appears indicating the alarm and profile settings for all alarms on all managed servers in the enterprise will be reset to their default values.

4 Click **OK**.

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Edit an Alarm

Edit an alarm to change any of the following:

- **Basic alarm settings** - These settings allow you to enable or disable the alarm, determine whether an audible sound accompanies the alarm, set the alarm priority, and specify the trigger delay and resend delay values.
- **Alarm Scheduling settings** - These settings allow you to schedule the specific days and hours in which the alarm is in effect.
- **Notification profiles assigned to the alarm** - The notification profiles assigned to an alarm determine who is notified by email, and which SNMP nodes receive SNMP traps, when an alarm is triggered.

You can edit an individual alarm or simultaneously edit multiple alarms. Separate procedures are provided in this section to explain each of these alarm editing procedures.

Note the following about simultaneously editing multiple alarms:

- You have the option to select specific alarms, or to select all of the alarms on one page of the Configure Alarms screen
- You can edit only those settings that are common to all of the alarms you select for editing
- If you change a setting, the new setting applies to every alarm you selected for editing

To edit multiple alarms simultaneously, see “Edit Multiple Alarms Simultaneously” on page 42.

Installation Tree Hierarchy and Alarm Editing

When editing an individual alarm, or editing multiple alarms simultaneously, it is important to understand how the Installations tree node that you select affects the alarm configuration.

You can configure settings for an alarm at the Enterprise, Site Group, Site, or Server node of the Installations tree. The configuration of the alarm can be inherited by the lower nodes of the Installations tree.

The configuration of the alarm can then be changed (or edited) at a lower node. This capability allows you to configure the same alarm to behave differently for different Site Groups, Sites, or Servers.

For example, you can select a Site Group node in the Installations tree and disable the “Proxy Detected” alarm so that this alarm is disabled for all managed servers in the Site Group. Then, you can select an individual managed server within the Site Group and enable the “Proxy Detected” alarm for only that specific server. In this configuration, the “Proxy Detected” alarm can be raised only by the one managed server in the Site Group. The alarm is disabled for all other servers in the Site Group.

Use this hierarchical approach to alarm editing when you need specific alarms to behave differently at specific nodes of the Installations tree.

To edit an individual alarm, or simultaneously edit multiple alarms, you need Edit Alarm Configuration privileges, scoped to the selected installation.

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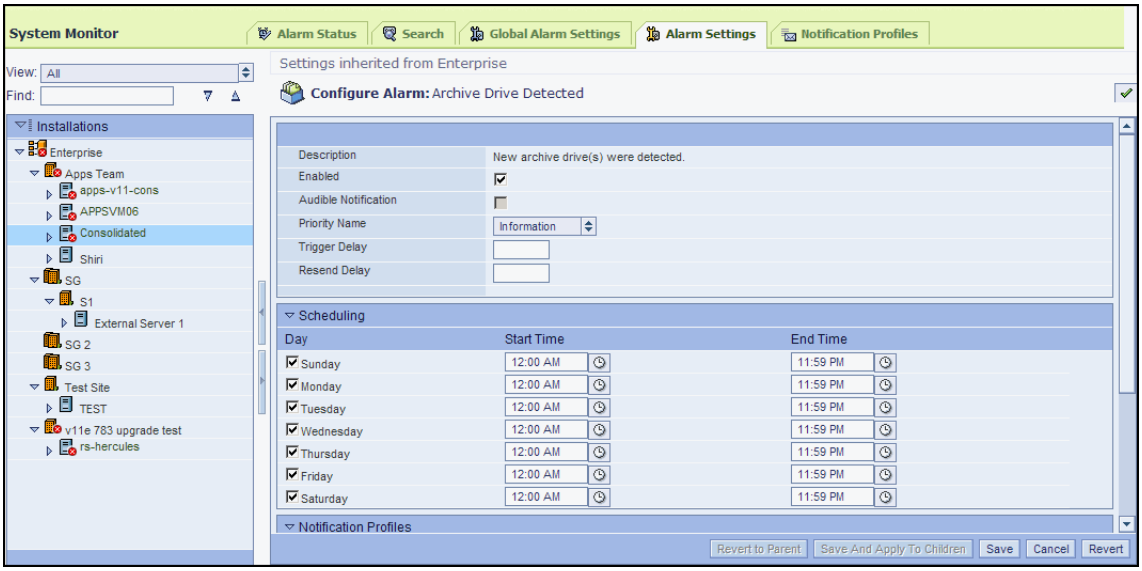
The procedures for editing an individual alarm, and simultaneously editing multiple alarms, are described separately below.

To edit an individual alarm:

1. Click **System Management** > **System Monitor** > **Alarm Settings**. The Configure Alarms screen displays a summary of all alarms as described in View or Reset Alarms on page 33.
2. In the Installations tree (left pane), select the node of the Installations tree from which you want to edit the alarm (Enterprise, Site Group, Site, or Server).
3. In the right-hand pane, select an individual alarm, and then click **Edit**. The Configure Alarm window displays.

NOTE You can also simultaneously edit multiple alarms. For more information, see "To edit multiple alarms simultaneously :" on page 43.



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4 Review and edit the fields according to the following:

Item	Description
Description	Shows a brief, read-only description of the alarm.
Enabled	Check to enable or disable the alarm. A checkmark indicates that the alarm is enabled (active). If the alarm is enabled, the alarm is triggered when the condition associated with it exists on a managed server. You can view the triggered (active) alarms as discussed in View a List of Active Alarms for a Server or a Server Role on page 17. When an alarm is disabled, the alarm is never triggered. This option is enabled by default. An asterisk (*) can appear in this field if you are simultaneously editing multiple alarms. The asterisk indicates that the alarms you are editing specify different values for this field. For more information, see "To edit multiple alarms simultaneously :" on page 43.
Audible Notification	Check to indicate that the alarm is to give off a warning sound when triggered. Clear the check mark to prevent the alarm from giving off a warning sound when triggered. If audible notifications are disabled in the Global Alarm Settings, this setting is greyed-out and cannot be enabled. For more information, see Configure Global Alarm Settings and Priorities on page 29. An asterisk (*) can appear in this field if you are simultaneously editing multiple alarms. The asterisk indicates that the alarms you are editing specify different values for this field. For more information, see "To edit multiple alarms simultaneously :" on page 43.
Priority Name	Select a priority level for this alarm. Default settings in order of severity from lowest to highest are **Information**, **Warning**, **Minor** and **Major**. You can change these names as well as add more alarm priorities, as described in Configure Global Alarm Settings and Priorities on page 29. An asterisk (*) can appear in this field if you are simultaneously editing multiple alarms. The asterisk indicates that the alarms you are editing specify different values for this field. For more information, see "To edit multiple alarms simultaneously :" on page 43.

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Item	Description
Trigger Delay	Specifies the alarm's Trigger Delay interval in seconds. This setting is disabled (either empty or set to zero) by default. In most cases, you should accept the default and leave this setting disabled. This is an advanced setting that is needed only in particular circumstances. This setting prevents administrators from being notified of alarms that require no action by the administrator to resolve. The Trigger Delay interval specifies the number of seconds that passes between the detection of an alarm and the time that the user is notified of the alarm (either through the appearance of the alarm in the user interface, an email notification, or an SNMP trap). Specify a Trigger Delay value only when a particular alarm repeatedly turns on and then turns off and requires no action by the administrator to resolve it. In some cases, the normal operating conditions in an environment may cause a specific alarm to be repeatedly turned on and off. For example, if your environment frequently experiences short network disconnections that trigger an alarm(s), configure this setting to prevent administrators from being repeatedly notified of these alarms. If you specify a value for this setting, the recommended value is 10 seconds. You may need to experiment with this value to determine the appropriate delay period. When this setting is disabled, there is no Trigger Delay and the alarm is raised as soon as it is detected. An asterisk (*) can appear in this field if you are simultaneously editing multiple alarms. The asterisk indicates that the alarms you are editing specify different values for this field. For more information, see "To edit multiple alarms simultaneously :" on page 43.
Resend Delay	Specifies the number of seconds that the alarm service waits to re-send an SNMP notification for an unacknowledged alarm. The alarm service sends an SNMP notification when it detects an alarm. This setting acts as a reminder notice, resending the SNMP notification at regular intervals until the alarm is acknowledged. The SNMP Notification option must be selected in the Global Alarm Settings tab for this option to be in effect. For more information, see "Configure Global Alarm Settings and Priorities" on page 29. This setting is blank by default. When this setting is blank, the SNMP notification is not re-sent. An asterisk (*) can appear in this field if you are simultaneously editing multiple alarms. The asterisk indicates that the alarms you are editing specify different values for this field. For more information, see "To edit multiple alarms simultaneously :" on page 43.

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Item	Description
Scheduling	Allows you to schedule the alarm. You can specify the days of the week, and the times of the day, that the alarm is in effect. Click the Clock icon to open the Time window and set times, or type times in the format hh:mm:AM/PM. The alarm can only be triggered during the times in which it is scheduled to be in effect. By default, an alarm is configured to be in effect seven days a week, 24 hours a day. An asterisk (*) can appear in this field if you are simultaneously editing multiple alarms. The asterisk indicates that the alarms you are editing specify different values for this field. For more information, see "To edit multiple alarms simultaneously :" on page 43.
Alarm when threshold is above/below *	This option does not appear for all alarms. If applicable, use this option to specify a specific value, or threshold, at which the alarm is triggered. Each alarm is different in that some alarms are triggered when the value rises *above* the specified threshold (threshold is above) and other alarms are triggered when the value falls *below* the specified threshold (threshold is below).
Acknowledge Alarm when threshold is above/below *	This option does not appear for all alarms. If applicable, use this option to specify a value, or threshold, at which the alarm is automatically acknowledged (that is, put to history/made inactive) by the system. Each alarm is different in that some alarms are automatically acknowledged when the value rises *above* the specified threshold (threshold is above) and other alarms are acknowledged when the value falls *below* the specified threshold (threshold is below). This option can work in combination with the **Alarm when threshold is above/below** option. For example, you can set the **Alarm when threshold is above** option to 3000 and the **Acknowledge Alarm when threshold is below** option to 2000. In this configuration, the alarm is raised (becomes active) when the value exceeds 3000 and is not automatically acknowledged by the system until the value goes back below 2000. In the example above, the alarm is not acknowledged until the value is safely below the threshold at which it was triggered. The two values should differ by a reasonable amount. Setting the two values too close together can cause an alarm to be repeatedly triggered and acknowledged by the system if the system wavers back and forth across the threshold value.

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Item	Description
Notification Profiles	To assign a notification profile to an alarm, select a profile from the **Available Profiles** pane (left) and click the right-pointing arrow to move that profile to the **Assigned Profiles** pane (right). Reverse this process to remove alarm notification profiles from this alarm. The notification profiles assigned to an alarm determine who is notified by email, and which SNMP nodes receive SNMP traps, when an alarm is triggered. See Alarms and the Pending Messages Icon on page 59 and "Work with Alarm Notification Profiles" on page 47 for more information. An asterisk (*) can appear in this field if you are simultaneously editing multiple alarms. The asterisk indicates that the alarms you are editing specify different values for this field. For more information, see "To edit multiple alarms simultaneously :" on page 43.
* These fields appear only for alarms that have thresholds. For alarms that do not have a threshold, these fields do not display.	

5 Do one of the following:

- Click **Save**. When you click **Save**, the changes are applied to the selected node. The changes are also applied to a child node if the child node has not been previously altered from its default state and saved.

  If the child node has been previously altered and saved, changes are not sent to the child node when you click **Save**. In this case, you must click the **Save and Apply to Children** button to apply the changes to the child node.
- Click **Save and Apply to Children** to copy the configured alarm settings to all managed servers or server roles under the selected Site Group, Site, or Server (that is, child nodes). Selecting this option sends a configuration message (cache update) to all of the affected Servers.
- Click **Revert to Parent** to ignore any changes. The Site Group, Site, Server or role (i.e. the node) then inherits the settings of its parent Server, Site, or Site Group.
- Click **Revert** if you begin making changes to an alarm and decide you want to start over with your changes before you have clicked the **Save** button. This button reverts the alarm settings to the settings that existed the last time the screen was saved.

Edit Multiple Alarms Simultaneously

If you want to change the same settings for several different alarms, you have the option to edit multiple alarms simultaneously.

The procedure is very similar to editing an individual alarm. Before editing multiple alarms simultaneously, you should be familiar with the information provided in "Edit an Alarm" on page 37.

Note the following about editing multiple alarms simultaneously:

- You have the option to select specific alarms, or to select all of the alarms on one page of the Configure Alarms screen.

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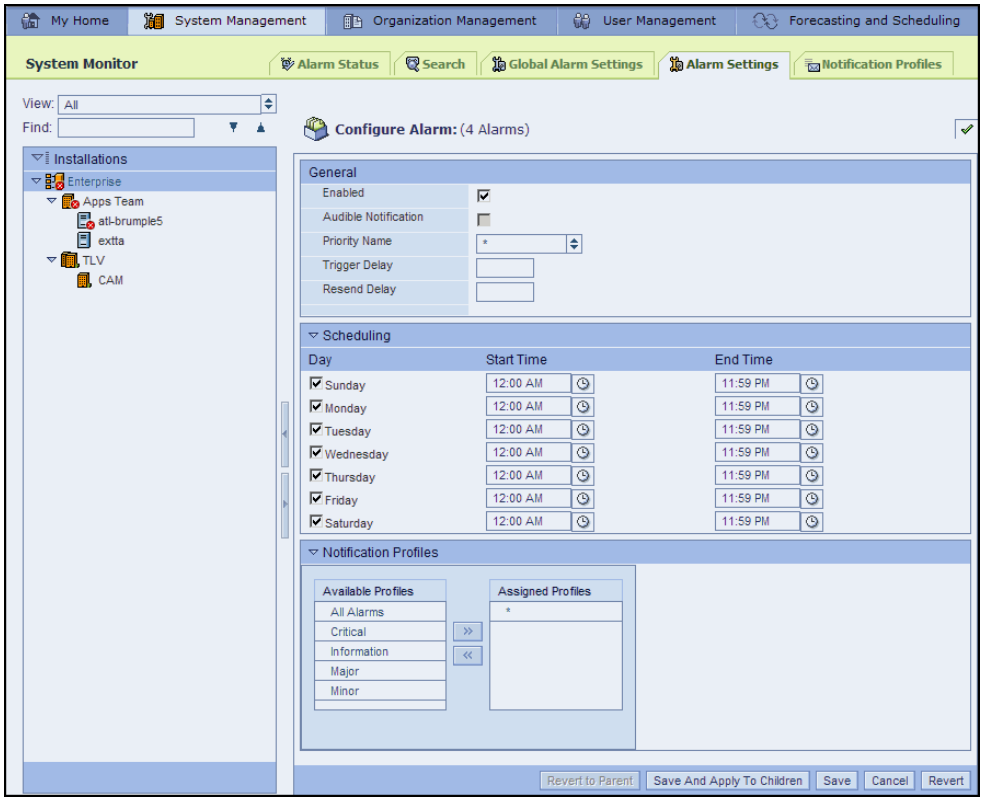
- You can edit only those settings that are common to all of the alarms you select for editing. Many alarms have identical settings. If all of the alarms you selected have identical settings, all alarm settings are available for editing. However, if one of the selected alarms has unique settings that other selected alarms do not have, you cannot edit those unique settings.
- If you change a setting, the new setting applies to every alarm you selected for editing.

To edit multiple alarms simultaneously :

Use this procedure if you want to simultaneously edit two or more alarms.

1. Click **System Management** > **System Monitor** > **Alarm Settings**. The Configure Alarms screen displays a summary of all alarms as described in View or Reset Alarms on page 33.
2. In the Installations tree (left pane), select the node of the Installations tree from which you want to edit the alarm (Enterprise, Site Group, Site, or Server).
3. In the right-hand pane, do one of the following:
   - Press and hold the **Ctrl** key while you select the specific alarms you want to edit, and then click **Edit**.
   - To edit all of the alarms on the currently displayed page, click the **Select All** button, and then click **Edit**.
4. The Edit Alarms screen displays.



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When simultaneously editing multiple alarms, note the following about the Edit Alarms screen:

- Only the alarm settings that are common to all of the alarms that you selected in step 3 display and are available for editing.

  For descriptions of every field that can appear on this screen, see "To edit an individual alarm:" on page 38.

- If an asterisk (*) appears as a value for a field, the alarms you have selected currently specify different values for this field.

  If you replace an asterisk with a specific value, all of the alarms you have selected will have the same value for the field after you complete this procedure.

  For example, assume you select three alarms for editing and two of the alarms have a Priority Name of Major, while one alarm has a Priority Name of Minor. In this case, an asterisk appears in the Priority Name field. If you replace the asterisk with the value Major, all three alarms will have Major as the Priority Name when you complete this procedure.

- If a specific value appears for a field (and not an asterisk) all the alarms you have selected currently have this same value specified for the field.

- Only the values you change on this screen will be applied to all of the alarms you selected in step 3 when you complete this procedure.

NOTE

A few of the alarms include an **Alarm when threshold is above/below** and an **Acknowledge alarm when threshold is above/below** field. These fields do not appear on the Edit Alarms screen when you select multiple alarms to edit. To change one of these fields, you must edit the alarm individually, as discussed in "To edit an individual alarm:" on page 38.

5 Change the values on the Edit Alarms screen as needed. Only the values that you change on the Edit Alarms screen will be applied to every alarm you selected in step 3.

6 Do one of the following:

- Click **Save**. When you click **Save**, the changes are applied to the selected node. The changes are also applied to a child node if the child node has not been previously altered from its default state and saved.

  If the child node has been previously altered and saved, changes are not sent to the child node when you click **Save**. In this case, you must click the **Save and Apply to Children** button to apply the changes to the child node.

- Click **Save and Apply to Children** to copy the configured alarm settings to all managed servers or server roles under the selected Site Group, Site, or Server (that is, child nodes). Selecting this option sends a configuration message (cache update) to all of the affected Servers.

- Click **Revert to Parent** to change all settings to be the same as the parent node of currently selected node. The parent Site Group, Site, Server or Server role (i.e. the node) then inherits, or reverts, to the settings of its parent Site Group, Site, or Server.

- Click **Revert** if you begin making changes to an alarm and decide you want to start over with your changes before you have clicked the **Save** button. This button reverts the alarm settings to the settings that existed the last time the screen was saved.

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Create, Edit, and Delete Alarm Filters

Alarm filters allow you to filter the alarms that display in the Configure Alarms screen available from the **Alarm Settings** tab.

The **All** alarm filter exists by default and displays all alarms associated with the node that is selected in the Installations tree.

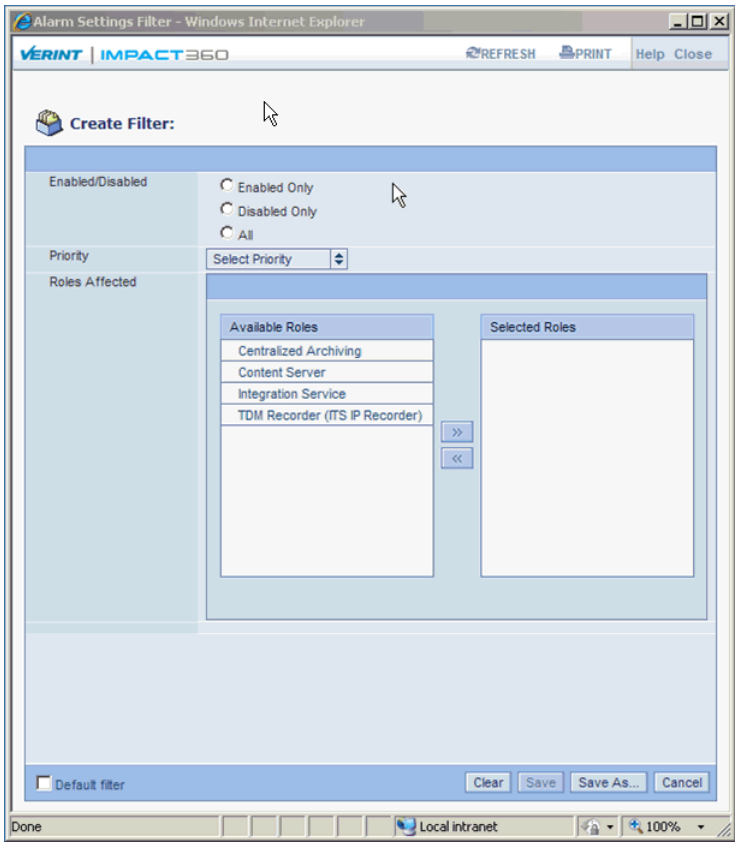
Create an alarm filter so that only alarms that meet the criteria you specify display in the Configure Alarms screen. Your three criteria for filtering alarms are Enabled/Disabled, Alarm Priority, and server role.

For example, you can create an alarm filter that displays only Enabled alarms that have the Major priority level and are associated with the Integration Service role.

You can also edit and delete alarm filters. You can edit or delete any alarm filter created by an administrator. You cannot edit or delete the **All** alarm filter, which is provided by default.

To create an alarm filter:

1. Click **System Management > System Monitor > Alarms Settings**.
2. Click the View menu in the upper-right of the screen, and then select **Create Filter**.



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3 Complete the fields according to the following:

Item	Description
Enabled/Disabled	Select one of the following to determine whether the alarm filter displays enabled alarms, disabled alarms, or both: • **Enabled Only** - Displays only alarms that are currently enabled. Enabled alarms are triggered whenever the condition associated with the alarm exists on a managed server. • **Disabled Only** - Displays only alarms that are currently disabled. Disabled alarms are never triggered. • **All** - Displays all alarms (both enabled and disabled).
Priority	Select a specific priority if you want the alarm filter to display only alarms of that priority. Choices include **Information**, **Warning**, **Major**, and **Minor**, or any other alarm priority that has been created in the global alarm settings, as described in Configure Global Alarm Settings and Priorities on page 29.
Roles Affected	Select specific server roles if you want the alarm filter to display only alarms that affect those server roles. To include a server role in the alarm filter, select a server role in the Available Roles pane and click the right-pointing arrow to move the role into the Selected Roles pane. To remove server roles from the alarm filter, click the server role in the Selected Roles pane, and then click the left-pointing arrow to move the role into the Available Roles pane.

4 Do one of the following:

- Click **Default Filter** to have this alarm filter appear as the default selection in the View area.
- Click **Clear** to clear all fields and start over from scratch.
- Click **Save As** to save the alarm filter as a new filter and provide it with a name. The new name then appears as a selection in the View menu.

To edit or delete an alarm filter:

1 Click **System Management > System Monitor > Alarms Settings**.

2 Click the View menu in the right-hand corner of the screen, and select the alarm filter you want to edit or delete.

NOTE You cannot delete the **All** alarm filter.

3 Click the View menu in the right-hand corner of the screen again and select **Edit Filter**.

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4 Edit the fields according to the following (skip this step if you are deleting the alarm filter):

Item	Description
Enabled/Disabled	Select one of the following to determine whether the alarm filter displays enabled alarms, disabled alarms, or both: • **Enabled Only** - Displays only alarms that are currently enabled. Enabled alarms are triggered whenever the condition associated with the alarm exists on a managed server. • **Disabled Only** - Displays only alarms that are currently disabled. Disabled alarms are never triggered. • **All** - Displays all alarms (both enabled and disabled).
Priority	Select a specific priority if you want the alarm filter to display only alarms of that priority. Choices include **Information**, **Warning**, **Major**, and **Minor**, or any other alarm priority that has been created in the global alarm settings, as described in Configure Global Alarm Settings and Priorities on page 29.
Servers with Roles	Select specific server roles if you want the alarm filter to display only alarms that affect those server roles. To include a server role in the alarm filter, select a server role in the Available Roles pane and click the right-pointing arrow to move the role into the Selected Roles pane. To remove server roles from the alarm filter, click the server role in the Selected Roles pane, and then click the left-pointing arrow to move the role into the Available Roles pane.

5 Do one of the following:

- Click **Default Filter** to have this filter appear as the default selection in the View area.
- Click **Clear** to clear all fields and make new selections for the filter being edited.
- Click **Save** to save your changes.
- Click **Save As** if you want to save the edited filter with a new name.
- Click **Delete** if you want to delete the filter.

Work with Alarm Notification Profiles

This chapter explains how to work with alarm notification profiles.

You use alarm notification profiles to provide notifications to individual employees and the Network Management System (NMS) when individual alarms are triggered. Alarm notifications are either e-mails or SNMP traps.

Alarm notification profiles are discussed in the following topics:

- Alarm Notification Profile Overview, page 48
- Example Workflow for Configuring Alarm Notification Profiles, page 50

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- View the List of Alarm Notification Profiles, page 52
- Edit an Alarm Notification Profile, page 54
- View the List of Alarm Notification Profiles, page 52
- Delete an Alarm Notification Profile, page 59

Alarm Notification Profile Overview

You can configure an alarm notification profile to send two kinds of notifications when an alarm is triggered:

- **E-Mails** - E-mails are sent to employee e-mail addresses or external e-mail addresses. These e-mails contain descriptive information about each triggered alarm.
- **SNMP traps** - SNMP traps are sent to SNMP nodes. The SNMP traps ensure a triggered alarm is reported to the Network Management System (NMS).

Essentially, an alarm notification profile is a list of e-mail addresses and/or SNMP nodes. Alarm notification profiles are assigned to alarms. When a specific alarm is triggered, employees listed in the notification profiles assigned to that alarm receive an e-mail, and SNMP nodes listed in the notification profile assigned to that alarm receive an SNMP trap.

There are two types of notification profiles:

- Pre-defined Notification Profiles, page 48 - Pre-defined profiles are available by default following a system installation.
- Custom Notification Profiles, page 49 - If the pre-defined notification profiles are not sufficient for your needs, you can create custom notification profiles.

Pre-defined Notification Profiles

Pre-defined notification profiles are the simplest to use. All individual alarms are configured by default to use the pre-defined notification profiles.

The five pre-defined notification profiles available following a system installation are described in the table below:

Notification Profile Name:	By default, this profile is assigned to:	An employee or SNMP node added to this notification profile is notified of:
All Alarms	Every alarm	Every alarm that is triggered.
Critical	Alarms of the Major priority	Every alarm of the Major priority that is triggered.
Major	Alarms of the Minor priority	Every alarm of the Minor priority that is triggered.
Minor	Alarms of the Warning priority	Every alarm of the Warning priority that is triggered.
Information	Alarms of the Information priority	Every alarm of the Information priority that is triggered.

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Pre-defined Notification Profile Usage Notes

By default, each alarm has two pre-defined notification profiles assigned to it:

- All Alarms
- One other notification profile depending on the Priority level of the alarm, as noted in the table above.

You add employee names and/or SNMP nodes to the pre-defined notification profiles to ensure that system administrators are notified when an alarm is triggered.

To illustrate how the notification profiles work, consider the example of the alarm named "A background job failed in Enterprise Manager."

This alarm has two alarm notification profiles assigned to it by default: the **All Alarms** profile and the **Critical** profile (this alarm has the Major priority).

When the "A background job failed in Enterprise Manager" alarm is triggered, every employee listed in the **All Alarms** profile and every employee listed in the **Critical** profile receives a notification e-mail. If the same employee is in both notification profiles, that employee receives two e-mails

Similarly, every SNMP node listed in the **All Alarms** profile and every SNMP node listed in the **Critical** profile receives an SNMP trap when the alarm is triggered.

IMPORTANT To make notification profiles operational, you must perform procedures in addition to adding employees and SNMP nodes to the notification profiles. To understand all the configurations needed to make notification profiles operational, see "Example Workflow for Configuring Alarm Notification Profiles" on page 50.

Custom Notification Profiles

If the pre-defined notification profiles are not sufficient for your enterprise, or you need to create a notification profile for a special purpose, you can create a custom notification profile.

To illustrate one way in which custom profiles can be used, assume that the following situation exists:

- There are two alarms of the Major priority that are triggered persistently.
- You want employees Mike Marshall and Alison Allard to be notified when these two alarms are triggered, but you do not want Mike and Alison to be notified of any other alarms.

In this situation, you can do the following:

- Create a new, custom notification profile.
- Add only employees Mike Marshall and Alison Allard to the custom profile.
- Assign the custom profile to the two alarms of the Major priority that are recurring.

Mike and Alison will each receive e-mails when either of the two alarms is triggered, but will not be notified of any other alarms if their names are not included in any other notification profiles.

The example above illustrates only one use of custom profiles. You can use custom profiles in any way that is suitable for your enterprise. The use of custom profiles in your enterprise is at your discretion.

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To create a custom notification profile, see "View the List of Alarm Notification Profiles" on page 52.

To assign an alarm notification profile to an alarm, see "Edit an Alarm" on page 37.

Example Workflow for Configuring Alarm Notification Profiles

This section provides an example of the complete workflow required to configure alarm notification profiles so that notification e-mails are sent to employees and SNMP traps are sent to SNMP nodes.

The example workflow is described in these topics:

- Notification Profile Configuration Prerequisites, page 50
- Notification Profile Configuration Workflow Example, page 50

Notification Profile Configuration Prerequisites

Before configuring alarm notification profiles ensure that the prerequisite conditions for supporting e-mail notifications and SNMP notifications are met.

- To support e-mail notifications for triggered alarms, you must first configure the system to work with an SMTP E-Mail server.

  Perform this configuration by completing the **SMTP Settings** on the **System Management > Enterprise Management > Enterprise Settings** tab. For details, see the "Enterprise Settings" section of the *Enterprise Manager Configuration and Administration Guide*.

- To support sending SNMP traps for triggered alarms, you must obtain and configure the system Management Information Base (MIB) files. For details about these configurations, see "" on page 64.

Alarm notification profiles are discussed in the following topics:

- Alarm Notification Profile Overview, page 48
- Example Workflow for Configuring Alarm Notification Profiles, page 50
- View the List of Alarm Notification Profiles, page 52
- Edit an Alarm Notification Profile, page 54
- View the List of Alarm Notification Profiles, page 52
- Create a Custom Alarm Notification Profile, page 59

Notification Profile Configuration Workflow Example

The steps below illustrate the process of setting up notification profiles to support e-mail and SNMP notifications for triggered alarms.

This example assumes you use the pre-defined notification profiles in your configuration, and describes the most basic configuration of notification profiles.

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Once you understand this process, you can vary from it if as necessary to customize the configuration for your enterprise. For example, you can create your own custom notification profiles, edit existing notification profiles at different nodes of the Installations tree, and so on.

To configure notification profiles that support e-mail and SNMP notifications:

1. Ensure the notification profile prerequisites are met as described in "Notification Profile Configuration Prerequisites" on page 50.
2. Add SNMP nodes to the system:
   a. Click **System Management > System Monitor > Global Alarm Settings**.
   b. In the **SNMP Node Settings** section, add every SNMP node that must receive an SNMP trap when an alarm is triggered. For details, see "Configure Global Alarm Settings and Priorities" on page 29.
3. Add employees and SNMP nodes to the five pre-defined notification profiles (All Alarms, Critical, Information, Major, and Minor). For example:
   a. Click **System Management > System Monitor > Notification Profiles**.
   b. In the left pane, select the Enterprise node.
   c. Select the **All Alarms** profile, and click **Edit**.
   d. In the **E-Mail Address** section, select an **Organization**, and then move the employees you want to be part of this notification profile into the **Selected Employees** section.

      For example, in the **All Alarms** profile, move the employees who must be notified of every alarm that is triggered into the **Selected Employees** section.
   e. In the **SNMP Nodes** section, move the SNMP nodes that you want to be part of this notification profile into the **Assigned SNMP nodes** section.

      For example, in the **All Alarms** profile, move the SNMP nodes that must receive traps when every alarm is triggered into the **Assigned SNMP nodes** section.
   f. Click **Save And Apply to Children**.
   g. Repeat steps b. through f. above for each remaining default notification profile (Critical, Information, Major, Minor).

      For example, in the Critical notification profile, specify all Employees and SNMP nodes to be notified if an alarm of the Major priority is triggered. In the Major profile, specify all Employees and SNMP nodes to be notified if an alarm of the Minor priority is triggered, and so on.
4. Review the notification profiles assigned to each alarm by default, and change the default assignments if necessary.

   NOTE In most cases, the default settings should be acceptable for the initial configuration of notification profiles. As you gain experience with alarms and notification profiles, you can alter the notification profile assignments as needed.

   a. Click **System Management > System Monitor > Alarm Settings**.

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b. In the left pane, select the Enterprise node.

c. In the right pane, examine the profiles assigned to the first alarm (listed under the **Profiles Assigned** column).

- If the profiles assigned to a specific alarm are acceptable, continue to the next alarm. (The assigned profiles determine the employees and SNMP nodes that are notified when the alarm is triggered.)
- If the profiles assigned to a specific alarm are not acceptable, select the alarm, click **Edit**, and change the notification profile assignments as needed. For more information, see "Edit an Alarm" on page 37.
- Continue performing step c. until you have the appropriate notification profiles assigned to each alarm in the system.

5 Enable E-mail notification and SNMP notification.

a. Click **System Management > System Monitor > Global Alarm Settings**

b. Select **Email Notification**.

c. Select **SNMP Notification**.

d. Click **Save**.

Notes About the Configuration Workflow Example

- In step 3, it is not mandatory to add employees and SNMP nodes to every pre-defined notification profile. System administrators should rely on their knowledge of their enterprise to determine whether all of the default notification profiles are needed, or only a subset of them. For example, some enterprises may want notifications to be sent only for alarms of the Major and Minor priorities. In this case, it is only necessary to add employees and/or SNMP nodes to the Critical and Major notification profiles.
- If the default notification profiles are not sufficient for your organization, you can create custom notification profiles as needed. See "Custom Notification Profiles" on page 49 and "View the List of Alarm Notification Profiles" on page 52 for more information.
- Once you have completed your basic notification profile configuration, review the information provided in "Edit an Alarm Notification Profile" on page 54 to achieve a more advanced configuration of the notification profiles. This section explains how editing notification profiles at different nodes of the Installations tree hierarchy affects notification profile behavior.

View the List of Alarm Notification Profiles

You can view a list of the alarm notification profiles that currently exist in the Enterprise Manager (both pre-defined profiles and custom profiles). The list of alarm notification profiles provides basic information about each alarm notification profile, such as the e-mail addresses and SNMP nodes assigned to the profile.

When viewing the list of notification profiles, the Installation tree node selected in the left-hand pane affects the notification profile details that appear in the right-hand pane. Notification profiles can be configured differently at each node of the Installations tree, as discussed in "Edit an Alarm Notification Profile" on page 54.

From this list you can also create new profiles, and edit or delete existing ones.

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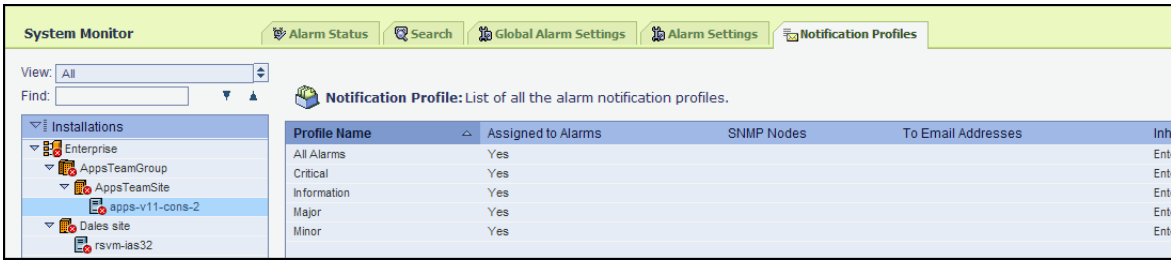
To view the list of alarm notification profiles:

1 Choose **System Management > System Monitor** > **Notification Profiles**.

2 In the Installations tree (left pane), select the Enterprise node. The list of all alarm notification profiles displays in the right-hand pane.

You can select lower nodes of the Installations tree (Site Group, Site, or Server) to display the notification profiles that are assigned to alarms at each node of the Installations tree.



3 The following information displays for each of the notification profiles:

Field	Description
Profile Name	Shows the names assigned to the alarm notification profiles. The five default notification profiles are All Alarms, Critical, Information, Major, and Minor.
Assigned to Alarm	Shows a Yes/No list indicating whether the notification profile is assigned to any alarms. By default, each of the pre-defined alarm notification profiles is assigned to multiple alarms. Pre-defined notification profiles are assigned to alarms based on each alarm's priority, as discussed in "Pre-defined Notification Profiles" on page 48. You can view the notification profiles assigned to alarms by selecting **System Management > System Monitor > Alarm Settings**.
SNMP Nodes	Shows the name of the SNMP (Simple Network Management Protocol) nodes that are assigned to this notification profile. These SNMP nodes are notified (sent an SNMP trap) when an alarm with this profile assigned is triggered. By default, no SNMP nodes are assigned to any of the notification profiles. To assign SNMP nodes to a notification profile, see "Example Workflow for Configuring Alarm Notification Profiles" on page 50 and "Delete an Alarm Notification Profile" on page 59.

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Field	Description
To E-Mail Addresses	Shows the e-mail addresses that are assigned to this alarm notification profile. Each e-mail address is separated by a comma or semicolon. There can be zero or many addresses listed here. These e-mail addresses are notified when an alarm associated with this profile is triggered. By default, no e-mail addresses are assigned to any of the notification profiles. To assign e-mail addresses to a notification profile, see "Example Workflow for Configuring Alarm Notification Profiles" on page 50 and "Delete an Alarm Notification Profile" on page 59.
Inherited From	Shows the node of the Installations tree from which the notification profile's configuration is inherited. Notification profiles can be configured differently at each node of the Installations tree, and the notification profile configuration can be inherited by the child nodes. For more information, see "Edit an Alarm Notification Profile" on page 54.

4 After viewing notification profiles, you can do any of the following:

- Click **Create** to create a new notification profile. See "View the List of Alarm Notification Profiles" on page 52.
- Click **Edit** to edit an existing notification profile. See "Edit an Alarm Notification Profile" on page 54.
- Click **Delete** to delete a notification profile. See "Delete an Alarm Notification Profile" on page 59.

Edit an Alarm Notification Profile

Edit an alarm notification profile to change the employees (e-mail recipients) or SNMP nodes that are notified by a profile when an alarm is triggered.

Installation Tree Hierarchy and Editing Notification Profiles

Before you edit a notification profile, it is important to understand how the Installations tree node that you select when editing a notification profile affects the operation of the notification profile.

You can configure (edit) a notification profile differently at each node of the Installations tree (Enterprise, Site Group, Site, or Server).

This capability allows you to configure a notification profile so that different employees (or SNMP nodes) receive notifications about alarms if the same alarm is triggered on different servers.

To illustrate, consider the following example scenario:

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- Assume you have two Sites in your enterprise: the Milan Site and the Rome Site.
- Servers in the Milan Site are administered by the Milan administrators group
- Servers in the Rome Site are administered by the Rome administrators group.
- You want the Milan administrators to be notified only of alarms on the Milan Site servers. You do not want Milan administrators to be notified of alarms on the Rome Site server.
- You want the Rome administrators to be notified only of alarms on the Rome Site servers. You do not want the Rome administrators to be notified of alarms on the Milan Site servers.

To achieve this configuration, you edit the notification profiles differently at the Milan and Rome Site nodes of the Installations tree.

For example:

1. Choose **System Management > System Monitor > Notification Profiles**.
2. In the left pane, select the Milan Site node.
3. With the Milan Site node selected in the left pane, select the All Alarms profile in the right pane.
4. In the All Alarms profile, do the following:
   - In the E-Mail Address section, move only the Milan administrators into the Selected Employees box.
   - In the SNMP Nodes section, move only the Milan SNMP nodes into the Assigned SNMP Nodes box.
   - Click Save and Apply to Children. (This applies the change to all of the server nodes in the Milan Site.)
5. In the left pane, select the Rome Site node.
6. With the Rome Site node selected in the left pane, select the All Alarms profile in the right pane.
7. In the All Alarms profile, do the following:
   - In the E-Mail Address section, move only the Rome administrators into the Selected Employees box.
   - In the SNMP Nodes section, move only the Rome SNMP nodes into the Assigned SNMP Nodes box.
   - Click Save and Apply to Children. (This applies the change to all of the server nodes in the Rome Site.)

In this example configuration, note that the All Alarms notification profile is configured differently at the Milan Site (see steps 3 and 4 above) than it is at the Rome Site (see steps 6 and 7 above).

Because the notification profile is configured differently at each Site, the Milan administrators are notified of all alarms occurring on the servers in the Milan Site, but none of the alarms occurring at the Rome Site. Similarly, the Rome administrators are notified of all alarms occurring on the server in the Rome Site, but none of the alarms occurring at the Milan Site.

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In this example, you would continue the configuration so that the remaining notification profiles (Critical, Major, Minor, Information) are also configured appropriately at the Milan and Rome Site nodes.

Note that, by default, all notification profiles inherit their configuration from the Enterprise node. You change the default behavior by editing a particular notification profile at a particular node of the Installations tree.

To edit an alarm notification profile:

1. Choose **System Management > System Monitor > Notification Profiles**.
2. Select the Installations tree node from which you want to edit the notification profile (Enterprise, Site Group, Site, or Server). The node that you select determines the managed servers to which the notification profile changes apply.
3. Click **Create**.
4. Edit the fields on the Notification Profile screen as needed. See "Notification Profile Screen Reference" on page 57 for details.
5. Do one of the following:
   - Click **Save**. When you click **Save**, the changes are applied to the selected node. The changes are also applied to a child node if the child node has not been previously altered from its default state and saved.

     If a child node has been previously altered and saved, changes are not applied to the child node when you click the **Save** button. In this case, you must select the **Save and Apply to Children** button to apply the changes to the child nodes.
   - Click **Revert to Parent** to ignore any changes. The Site Group, Site, or Server that is currently selected in the Installation tree then inherits the settings of its parent Site, or Site Group.
   - Click **Save and Apply to Children** to copy the configured notification to all Servers under the selected Site Group, Site, or Server (that is, child nodes).
   - Click **Revert** to revert to the settings that existed before you started this procedure (the settings revert to the settings that existed the last time the screen was saved).

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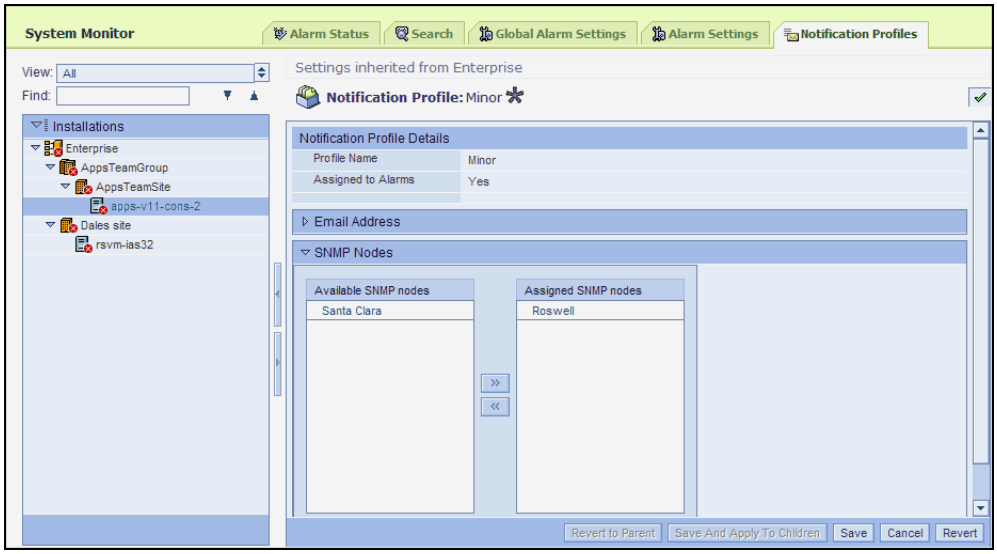
Notification Profile Screen Reference

Complete or change the options on the Notification Profile screen when creating or editing a notification profile.



The options on the Notification Profile screen include the following:

Field	Description
Notification Profile Details	
Profile Name	Type the name you want to assign to the alarm notification profile.
Assigned to Alarms	Read-only field that indicates if the Profile Name has been assigned to any alarms.

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Field	Description
Email Address	
Employees - Organization	If you are adding employees to the notification profile, select the Organization that contains the employees you want to add. All employees in the selected Organization will display in the Available Employees window.
Available Employees	Shows a list of employees belonging to the selected Organization. To assign an employee's e-mail address to this notification profile, select the employee's name and click the right-pointing arrow to move the employee's name into the Selected Employees box.
Selected Employees	Shows a list of every employee currently assigned to the notification profile. These employees receive notification e-mails if this notification profile is assigned to an alarm and that alarm is triggered. The e-mail is sent to the e-mail address specified in each employee's user profile. To remove an employee's e-mail address from this notification profile, select the employee's name in the Selected Employees box and click the left-pointing arrow to move the employee's name into the Available Employees box.
Additional Email Addresses	Type other e-mail addresses you want to associate with the notification profile. You must separate multiple entries with a comma or semicolon. There can be many addresses listed here or the field can be left blank. When an alarm associated with this notification profile is triggered, the Enterprise Manager also sends an e-mail notification to these e-mail addresses. Typically, e-mail addresses listed here are not available in an employee's user profile. For example, some users may want personal e-mail addresses listed here so that they also receive alarm notification e-mails at their personal e-mail addresses.
SNMP Nodes	
Available SNMP nodes	Shows a list of all SNMP servers/nodes that are specified in the Global Alarm Settings. (To make SNMP nodes available in this list, see Configure Global Alarm Settings and Priorities, page 29.) To assign an available SNMP node to this notification profile, select it and then click the right-pointing arrow to move it into the Assigned SNMP nodes box. When an alarm associated with this notification profile is triggered, an SNMP trap is sent to this SNMP node.
Assigned SNMP nodes	Shows the SNMP node(s) that have been assigned to the current notification profile. These SNMP nodes receive SNMP traps when an alarm to which this notification profile is assigned is triggered. To remove an Assigned SNMP node from this notification profile, select it and then click the left-pointing arrow to move it into the Available SNMP nodes box.

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Create a Custom Alarm Notification Profile

Create a new, custom alarm notification profile if the pre-defined alarm notification profiles are not sufficient for your enterprise. After you create a new alarm notification profile, you assign it to an alarm as described in Edit an Alarm, page 37.

You can create a notification profile only from the Enterprise node.

To create a new alarm notification profile:

1. Choose **System Management > System Monitor > Notification Profiles**.
2. In the Installations tree (left pane) select the Enterprise node.
3. Click **Create**.
4. Complete the fields on the Notification Profile screen. See "Notification Profile Screen Reference" on page 57 for details.
5. Click **Save**.
6. To assign the new profile to alarms, see "Edit an Alarm" on page 37.

Delete an Alarm Notification Profile

Delete a profile to remove the alarm notification profile from the list of alarm notification profiles and to unassign the profile from all alarms.

You can only delete alarm notification profiles from the Enterprise node of the Installation tree.

To delete an alarm notification profile:

1. Choose **System Management > System Monitor > Notification Profiles**.
2. In the Installations tree (left pane), select the Enterprise node. The list of alarm notification profiles displays in the right-hand pane.
3. In the **Notification Profile** window, select the profile to be deleted.
4. Click **Delete**. A warning displays if the profile is associated to any alarms.
5. Click **OK** to confirm the deletion. The profile is deleted and a cache update is sent to all managed servers that had alarms associated to the profile.

Alarms and the Pending Messages Icon

The Pending Messages icon can appear on any screen accessible from the **System Management > System Monitor** menu from which you make configuration changes. The Pending Messages icon appears just to the right of the screen title above the right-hand pane.

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The Pending Messages icon indicates that configuration changes were made on a screen but the changes have not yet been sent to the managed server. For more information about this icon, see "The Pending Messages Icon" on page 142.

View Detailed Status of a Managed Server

From Enterprise Manager, you can launch the Enterprise Manager Agent (EMA) application or the Recorder Manager (RM) application on a managed server and use the System Monitor on the EMA or RM to view detailed status information about the managed server.

The EMA or RM System Monitor displays the managed server's status using a Metrics list and/or a Status Messages list.

The Metrics list provides a list of components used by a managed server or a server role. Specific metrics (such as "Memory Used" or "Active Alarm Count") are associated with each component. A color-coded status icon or bar indicates the status of the condition measured by the metric as noted below:

- Red indicates an error (a condition requiring attention)
- Orange indicates a warning status (a condition approaching an error state)
- Green indicates a good status
- Grey indicates an unknown status

The Status Messages list also displays color-coded status icons (red, orange, green or grey). A text message associated with each status icon provides details on the status indicated by the icon.

The list that displays (Metrics list or Status Messages list) depends on the server roles that are active on the managed server. Some server roles display only the Metrics list, some server roles display only the Status Messages list, and other server roles display both the Metrics list and the Status Messages list.

If you select a managed server, the Metrics and/or Status Messages lists for all server roles on the server display. If you select a specific server role, only the Metrics and/or Status Messages list associated with that server role display.

To view the status metrics for a Server or a Server role:

1. Click **System Management > System Monitor > Alarm Status**.
2. In the Installations tree (left pane), select the Server node for which you want to view status.
3. In the right-hand pane, either click on the row for the managed server and then click the **View** button, or double-click on the row for the server.

   Performing this step launches the Enterprise Manager Agent (EMA) application or the Recorder Manager (RM) application on the managed server. The EMA or RM includes a System Monitor feature that reports the status of the managed server.

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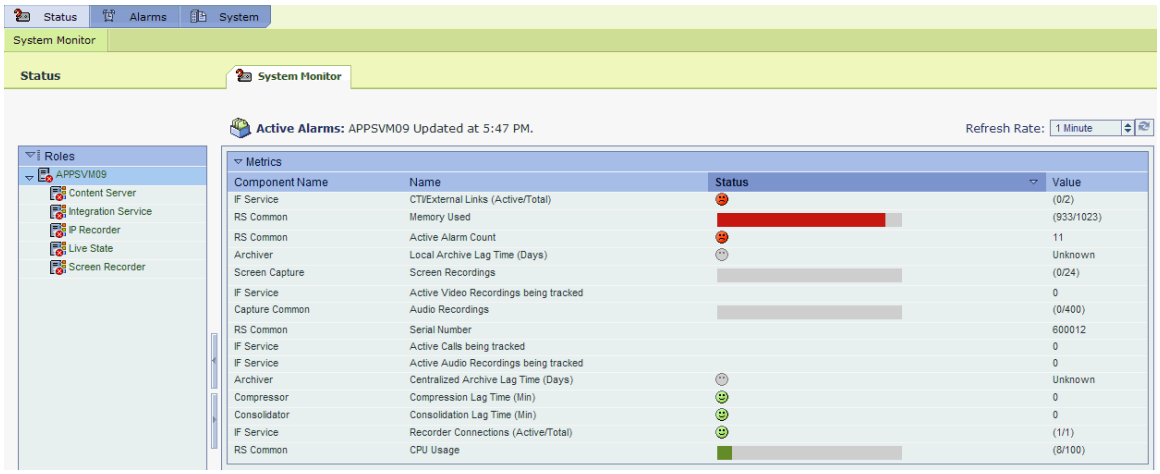
4 Click **Status > System Monitor**.



5 Review managed servers and their server roles. In the Roles (left-hand) pane, notice that:

- Clicking the arrow beside the Server icon displays and hides roles.
- A red error icon displaying on the Server or on a Server Role icon indicates the server has a condition that requires attention. This red icon displays on a Server icon if any one of its server roles have a metric with a red status.
- When you select a Server icon, the Metrics list and/or Status Messages list for all roles on the managed server display in the right-hand pane.
- When you select a specific server role, only the Metrics list and/or Status Messages list for that role display in the right-hand pane.

6 Click the Server or Server Role icon to view the Metrics list and/or Status Messages list associated with that managed server or server role.

To edit the thresholds at which some metrics display either a warning or an error, you must select the Server icon.

7 Review fields according to the following:

Item	Description
Metrics Section (This section does not appear for some server roles.)	
Component Name	The name of the component that provides status for the metric. (Each server role is comprised of one or more components.)
Name	The name of the metric for which status is reported.

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Item	Description
Status	Shows the status by color (red, orange, green, or grey). • Red status (or unhappy face) indicates the condition reported by the metric requires attention. • Orange status indicates a condition in a warning state. • Green status (or happy face) indicates the status is acceptable. • Grey status (or expressionless face) indicates that the status is unknown or the status is not updated. For some metrics (such as "Memory Used" or "CPU Usage") a status bar displays to indicate a percentage, or a specific quantity, of the resource that is used. The color of the status bar indicates whether the condition reported by the metric requires attention (red), is in a warning state (orange), is acceptable (green), or has no status (grey).
Value	The value associated with the metric (for example, the percentage of CPU usage, the specific number or alarms active on the server, or the specific number of active calls being tracked).
Status Messages Section (This section does not appear for some server roles.)	
Status	Shows the status by color (red, orange, green, or grey). • Red status (or unhappy face) indicates the condition reported by the metric requires attention. • Orange status indicates a condition in a warning state. • Green status (or happy face) indicates the status is acceptable. • Grey status (or expressionless face) indicates that the status is unknown or the status is not updated.
Message Details	Displays the status message text.
Refresh Rate	
Refresh Rate	Shows the current setting for refresh rate in minutes. Choose 1,2,3,4,5,10,15, or 20 minutes as the automatic refresh rate. At these times, the application communicates with the managed server and provides the Enterprise Manager with alarm updates. The default is 1 minute. You can also click the **Refresh** icon (inverted arrows beside the selection box) to update alarms immediately, without waiting for the automatic refresh cycle.

8 Do one of the following:

- Click **Edit Thresholds** to edit system monitor threshold values, as described in "Edit System Monitor Thresholds" on page 63. (To edit the thresholds, you must select the Server node in step 3.)
- Click the **Update** button to see the up-to-the-second status displayed in the Metrics list and/or the Text Status list.

NOTE The system may display status that is one or two minutes old, depending on the system performance. Click the **Update** button to ensure that you see the status as it exists at that instant.

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Edit System Monitor Thresholds

Edit the threshold values of EMA system monitor status fields to specify the thresholds at which a problem condition is reported as a warning or an error.

You can specify these thresholds:

- **Warning Threshold** - When this threshold is exceeded for a field, the icon or status bar associated with the field on the System Monitor Status screen turns orange.
- **Error Threshold** - When this threshold is exceeded for a field, the icon or status bar associated with the field turns red.

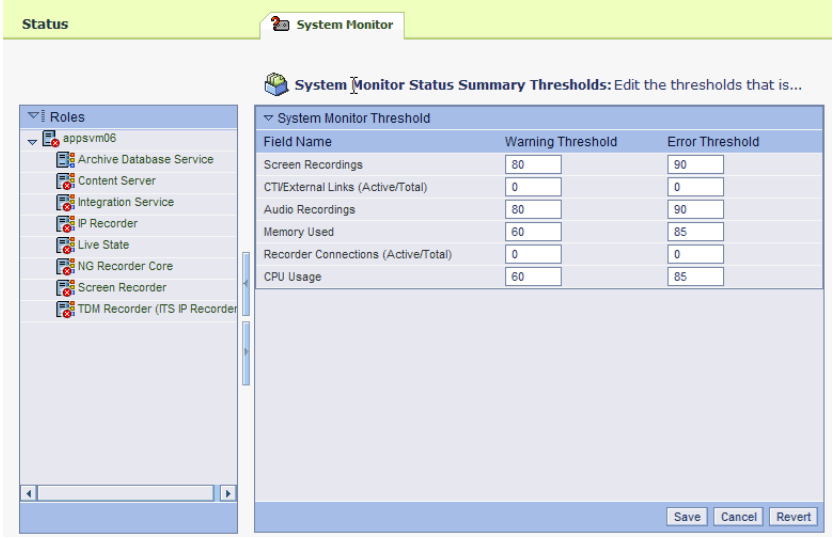
For example, assume you set the Warning and Error Thresholds for CPU Usage to 60 and 85 respectively. When CPU Usage reaches 60 percent, the status bar turns orange to indicate a warning status. When CPU Usage reaches 85 percent, the status bar turns red to indicate an error status.

The specific thresholds that are available on a server depend on the server roles that are activated on the server. Every server does not display the same set of thresholds.

The fields that display either have a predefined maximum value or can be configured as a percentage.

To edit System Monitor thresholds:

1 Click **Status > System Monitor**.



2 Click **Edit Thresholds**.

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3 In the System Monitor Thresholds window, do one of the following:

- Click **Default**, and then click **Save** to set the default values for all fields (recommended).
- Enter a number between one and 100 for the **Warning Threshold** and **Error Threshold** values for a field(s) and then click **Save**:

  For **Warning Threshold**, type a number that indicates the point at which a warning (orange icon or orange status bar) displays in the System Monitor - when the field of interest reaches this percentage or number, a warning message appears. The default values are recommended.

  For **Error Threshold**, type a number that indicates the point at which a warning displays in the System Monitor - when the field of interest reaches this percentage or number, an error (red icon or red status bar) displays. The default values are recommended.

  The specific values that appear in the Edit Threshold screen are dynamic. The values that display here are dependent on the specific functionality provided by the server. The table below provides an example of the thresholds that appear for a Recorder server.

Field	Threshold Default Value
Screen Recordings (%) This field indicates a percentage of the licensed recording capacity currently being used. For example, if your enterprise is licensed for 200 screen channels, and 160 screens are currently being recorded, the Screen Recordings percent is 80.	Warning Threshold default 80%.
	Error Threshold default 95%.
Audio Recordings (%) This field indicates a percentage of the licensed recording capacity currently being used. For example, if your enterprise is licensed for 200 voice channels, and 160 calls are currently being recorded, the Audio Recordings percent is 80.	Warning Threshold default 80%.
	Error Threshold default 90%.
Memory Used (%)	Warning Threshold default 60%.
	Error Threshold default 85%.
CPU Usage (%)	Warning Threshold default 60%.
	Error Threshold default 85%.

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Chapter 3

Working with SNMP MIB Files

You can configure alarms so that SNMP traps are sent to your Network Management System (NMS) when alarms are triggered on the managed servers. For information about configuring alarms so that SNMP traps are sent to your NMS when the alarms are triggered, see the topics “Work with Alarm Notification Profiles” and “Edit an Alarm” in Chapter 2.

SNMPv1, SNMPv2, and SNMPv2c are supported. SNMPv3 or higher versions are not supported.

In addition to performing the SNMP-related configurations noted above, you must also configure the Management Information Base (MIB) files so the your NMS can interpret the traps sent to it by the managed servers.

This chapter explains how to work with the SNMP MIB files and includes the following sections:

- Deploy the MIB Files, page 66 - Explains how to obtain and configure the MIB files.
- SNMP MIB Structure, page 75 - Provides reference information concerning the structure of the alarm trap.

Deploy the MIB Files

The MIB files allow network management systems to monitor any device defined within the Recording system.

Specifically, Management Information Base (MIB) files enable the SNMP manager to interpret trap messages sent from a managed server to the SNMP manager. To interpret a trap that the managed server sends, the SNMP manager must have access to the MIB that describes the format and content of the possible traps that the managed server can send.

IMPORTANT You use an SNMP configuration tool to configure the managed server MIB files. There are many different SNMP configuration tools on the market.

To illustrate the process of configuring the managed Server MIB files, this documentation assumes you are using the Castle Rock SNMPc Network Manager. If you are using a different SNMP configuration tool in your environment, you must vary from these procedures as appropriate for the tool you are using. Consult the documentation provided with your SNMP configuration tool as needed.

To deploy the five MIB files, you will need to:

- Obtain the MIB ZIP File
- Extract and Copy the MIB files
- Compile the MIB files
- Configure the Event Filter

Obtain the MIB ZIP File

The MIB files needed to monitor devices defined within the Recording system are included in the **V11 MIB Files.zip** file.

You can download the **V11 MIB Files.zip** file from the Internet, as described below:

1. Browse to the **Verint Online for Partners/Customers** web site (**https://ve5.verint.com**).
2. Enter the appropriate username and password.
3. From **My Communities**, select **Impact 360 V11 Partners**.
4. On the left, under **Support & Downloads**, select **Tools**.
5. Go to **Impact 360 V11 Tools > MIB Files > V11 MIB Files.zip**.
6. Save the **V11 MIB Files.zip** file to either your local system or an appropriate network location.

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Extract and Copy the MIB Files

After you have obtained the **V11 MIB Files.zip** file, extract the five MIB files it contains and copy them to the appropriate folder running your SNMP configuration tool.

The **V11 MIB Files.zip** file contains these five MIB files:

- WITS-CSFAMILY-MIB.mib
- WITS-CS-MIB.mib
- WITS-PRODUCTS-MIB.mib
- WITS-TRAP-MIB.mib
- WITS-TRAP-MIBv1.mib

After extracting these five MIB files from the zip file, copy them to the appropriate folder on the server running your SNMP configuration tool.

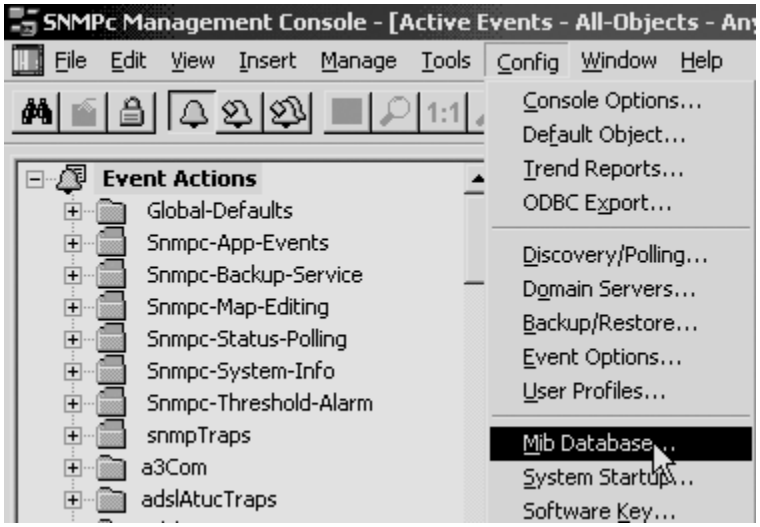
For example, if you are using the Castle Rock SNMPc Network Manager, copy the MIB files to **<Install Path>\SNMPc Network Manager\mibfiles** on the server running the Castle Rock SNMPc Network Manager.

Compile the MIB files

Use your SNMP configuration tool to compile the MIB files. The following section assumes you are using the Castle Rock SNMPc Network Manager. If you are using a different SNMP configuration tool in your environment, use these procedures as guidelines as you follow the specifc instructions in the tool's documentation.

1 Open **Start > Programs > SNMPc Network Manager > Startup System**.

2 In the SNMPc management console, choose **Config > MIB** database.



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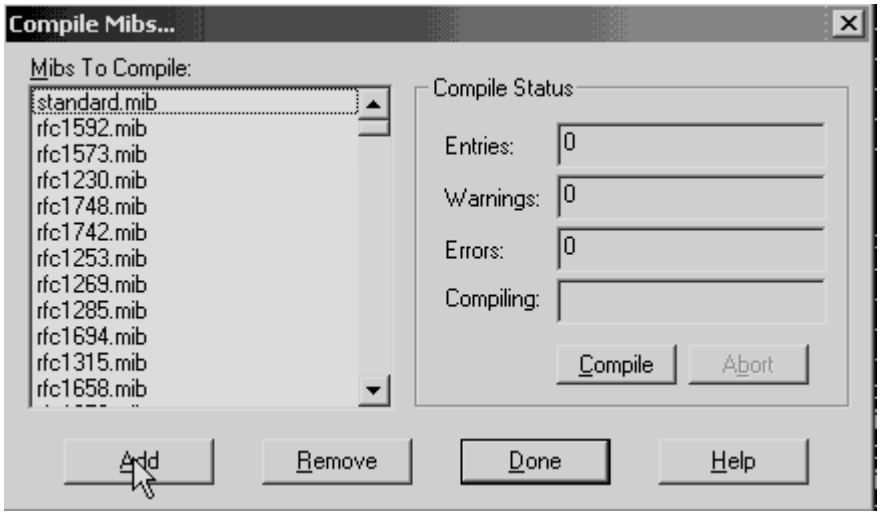
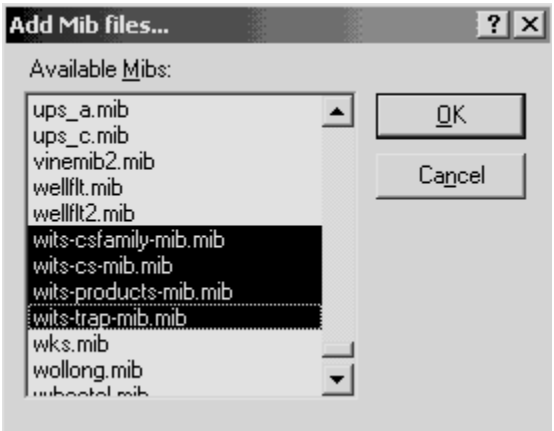
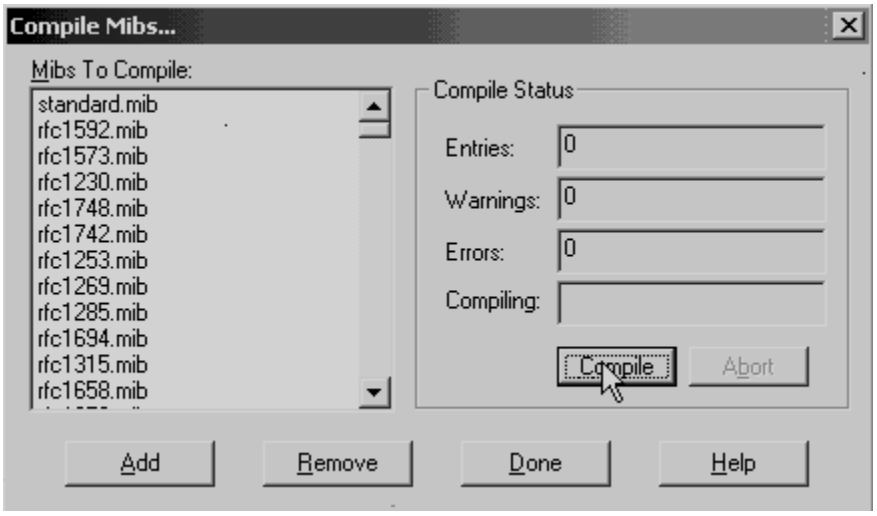
3 Click **Add** to select the Recorder MIB files.



4 Select the Recorder MIB files to add.



5 Click **Compile**. It will compile all the MIB files loaded.



6 Click **Done** to complete the compilation.

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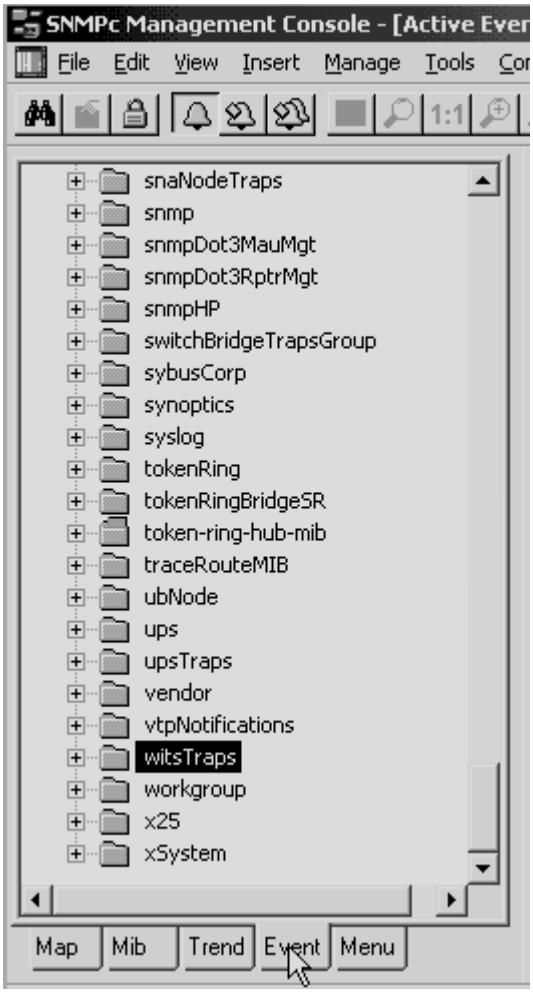
Configure the Event Filter

Once the files are compiled, set up the Event Filter. You need to set up a filter for each severity: Major, Minor, Information and Warning. The following steps show how to set up the Major filter.

1 In the left pane of the SNMPc Management Console, select the **Event** tab and navigate to **witsTraps**.



2 Expand the **witsTraps** folder.

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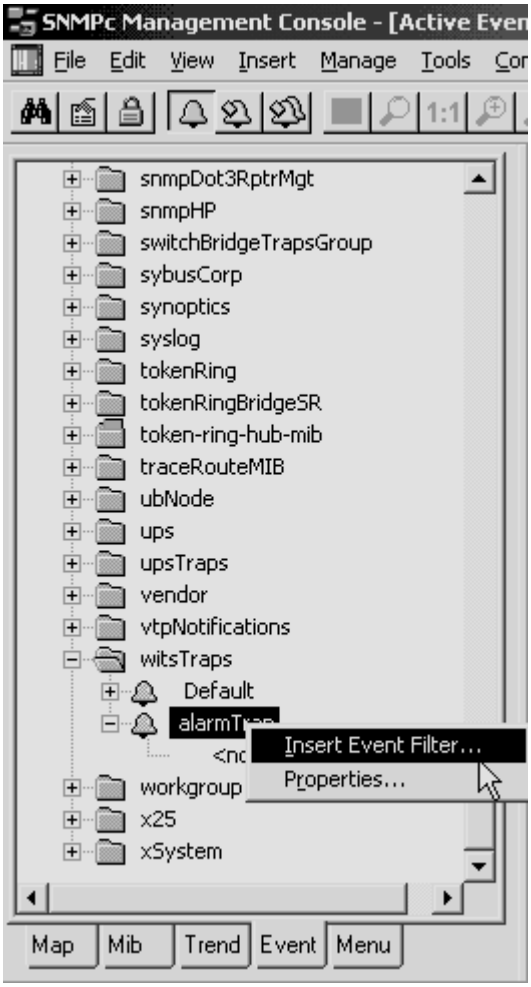
3 Navigate to **alarmTrap**.



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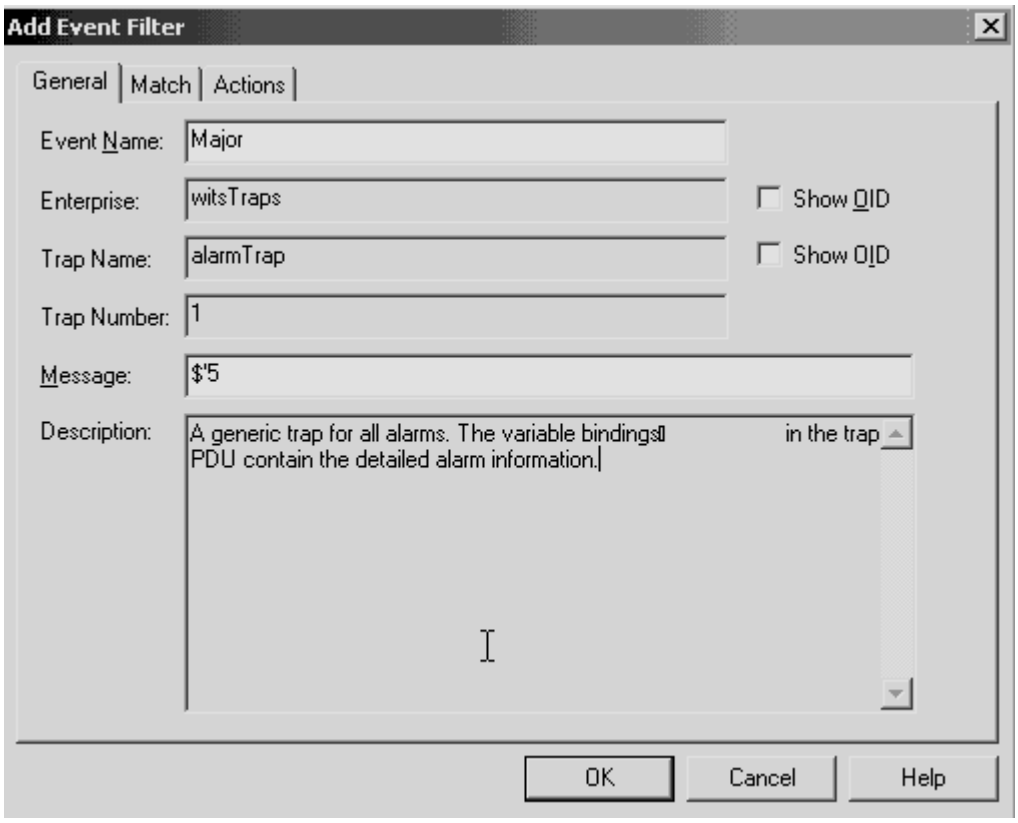
4 Right-click and then select **Insert Event filter...**



5 Within the **Add Event Filter** dialog, on the **General** tab:

a. Set **Event Name** as **Major**.

b. Set **Message** as **$'5**.

Setting message to $'5 extracts the message in field 5 of SNMP trap and displays it in the monitor window. Field 5 is the body of the alarm message, so this causes the alarm message to be displayed in the monitor window.

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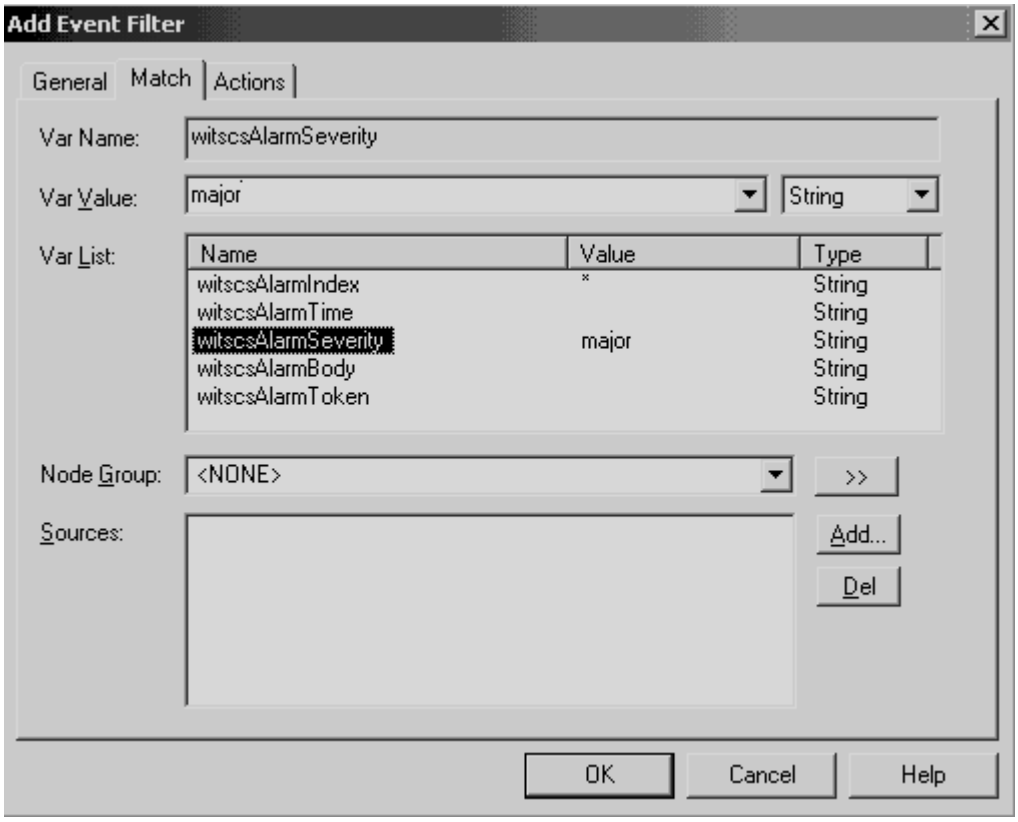
6 Click the **Match** tab.



7 Under the **Match** tab:

a. In the **Var List**, select **witscsAlarmSeverity**.

b. In the **Var Value**, select **Major** from the drop-down list.

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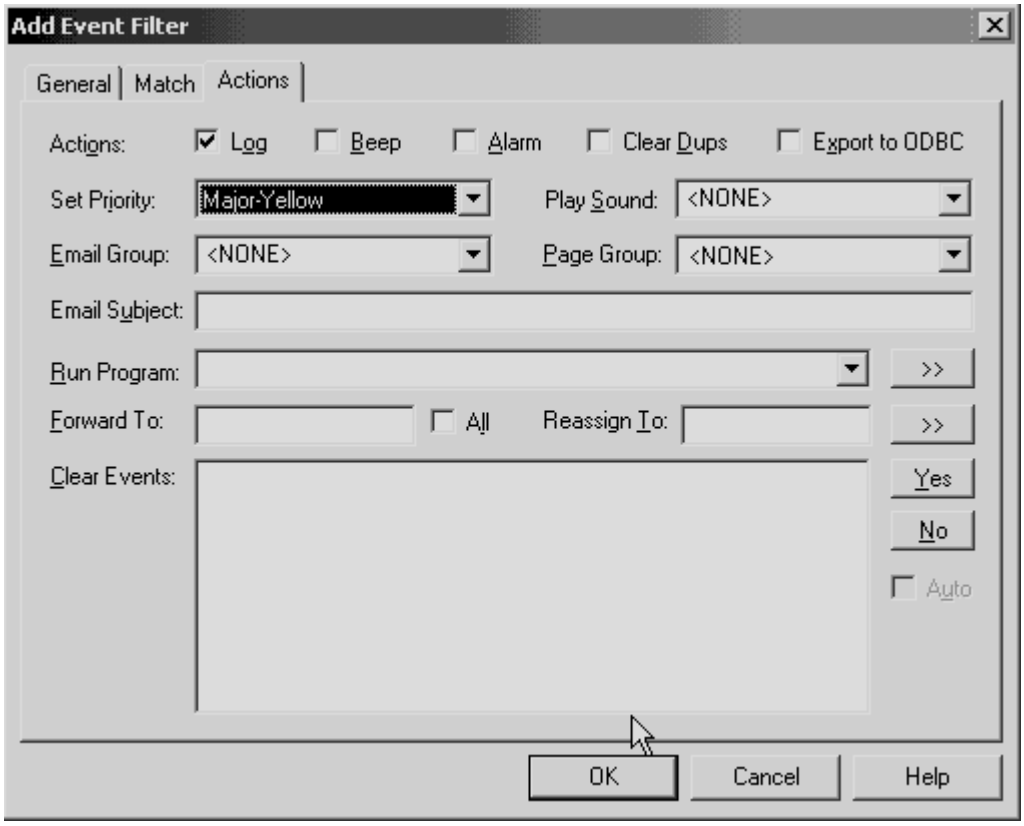
8 Under the **Actions** tab, set the **Priority** as **Major-Yellow** from the drop-down box.



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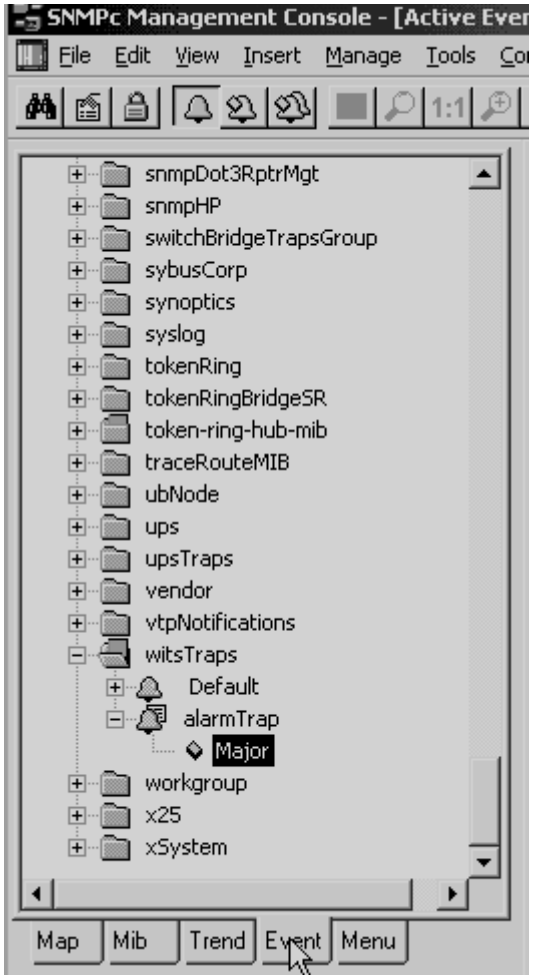
9 Click **OK** to complete the event filter configuration.

When configured properly, you will see a new **Event Filter** called **Major** under **Alarm Trap**.



10 Repeat these steps for all other severities; Minor, Information and Warning.

As you define each filter, assign the proper **Event Name** on the **General** tab (Minor, Information and Warning), the correct **Var Value** on the **Match** tab (Minor, Information and Warning), and the appropriate **Priority** on the **Actions** tab.

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SNMP MIB Structure

The SNMP MIB structure consists of a single alarm trap ID (OID = 1.3.6.1.4.1.1815.1.1.1.6.6.2) that includes the parameters listed in the following table. These MIB parameters are updated in the third-party SNMP monitoring application used in your environment.

Parameter	Object Identifier (OID)	Description
AlarmIndex	1.3.6.1.4.1.1815.1.1.1.6.6.2.1.1	A unique identifier of the alarm instance.
AlarmTime	1.3.6.1.4.1.1815.1.1.1.6.6.2.1.2	The time the alarm was raised or acknowledged.
AlarmSeverity	1.3.6.1.4.1.1815.1.1.1.6.6.2.1.3	The alarm severity: 0 - Informational 1 - Warning 2 - Minor 3 - Major
AlarmBody	1.3.6.1.4.1.1815.1.1.1.6.6.2.1.4	The message text for the alarm including the actual parameters (AlarmPx).
AlarmToken	1.3.6.1.4.1.1815.1.1.1.6.6.2.1.5	A unique predefined alphanumeric identifier of the alarm.
AlarmNumber	1.3.6.1.4.1.1815.1.1.1.6.6.2.1.6	A predefined numeric identifier for the alarm.
AlarmActive	1.3.6.1.4.1.1815.1.1.1.6.6.2.1.7	A flag that indicates whether the alarm has been acknowledged or is currently in the raised state: 0 - Acknowledged 1 - Raised
AlarmClientId	1.3.6.1.4.1.1815.1.1.1.6.6.2.1.8	The SNMP ClientId as set by the Enterprise Manager in the AlarmConfig.xml files: <snmpclientid> element
Common Data		
SysUpTime	1.3.6.1.4.1.1815.1.1.1.4.1	The number of tick counts for which the system has been running.

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Chapter 4

Understanding the Alarm Spreadsheet

Every alarm that can be raised by the system is listed in the *Alarms Spreadsheet*. The *Alarms Spreadsheet* is an Excel file that is provided as an attachment to the PDF version of this document. The *Alarms Spreadsheet* is also available as a separate standalone document.

NOTE To download the *Alarms Spreadsheet* from the PDF version of this document, click on the Attachment icon at the left edge of the Adobe PDF Reader application.

This chapter discusses the following topics to help you work with the *Alarms Spreadsheet*:

- Alarms Spreadsheet Contents, page 77
- Alarm Attributes, page 77
- Advanced Corrective Actions Are For Certified Support Personnel Only, page 80

Alarms Spreadsheet Contents

The *Alarms Spreadsheet* lists every alarm the system can raise, and also shows the default value of every attribute of every alarm. This document is useful in any alarm configuration activity, as you can quickly locate any alarm, or sort the entire list of alarms based on any alarm attribute (for example, you can sort alarms alphabetically by **alarm name** or numerically by **alarm number**).

The *Alarms Spreadsheet* includes these three tabs:

- **Start** - This tab contains introductory information for the Alarms Spreadsheet.
- **List of All Alarms** - This tab contains the alarms list. One alarm is listed on each row. The columns display the attributes of each alarm, and show the default values for each alarm. If an attribute value is empty for an alarm, that attribute is not applicable to the alarm.
- **Alarm Attribute Help** - This tab provides an explanation of each alarm attribute (that is, each column) shown in the **List of All Alarms** tab.

Alarm Attributes

In the *Alarms Spreadsheet*, one alarm is listed in each row of the spreadsheet. The spreadsheet columns list the attributes of each alarm.

A description of each alarm attribute (the column headings in the *Alarm Spreadsheet*) is provided below. This same information also appears on the **Alarm Attribute Help** tab of the *Alarms Spreadsheet*.

The alarm attributes include:

- **Component:** Shows the component to which an alarm belongs.

  Note that many alarms can be raised multiple components (that is, an alarm can be raised by a component other than the one to which it belongs).

  If you are viewing alarms in the user interface, the Active Alarms screen shows the component that raised the alarm, not the component to which the alarm belongs.
- **Alarm Name:** The alarm name as it displays in the user interface.
- **Internal Alarm Name:** The internal name for the alarm. This name appears in logs and is referenced by the software code, but does not display in the user interface.
- **Alarm Number:** A unique numerical identifier for the alarm. Some alarms have the same Alarm Name, but different alarm numbers.
- **Event Alarm:** Indicates whether the alarm is, or is not, triggered by a transient event. Alarms triggered by transient events indicate that a problem has occurred, but the problem often cannot be resolved by an administrator. This value can be On or Off.

  The **On** value indicates that the alarm is triggered by a transient event. The alarm only alerts the administrator that a problem has occurred but often the administrator cannot resolve the problem that the alarm reports. For example, a missed network packet may trigger an event alarm. In this case, the event

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happened in the past and the administrator cannot recover the lost packet. However, the alarm alerts the administrator that the event occurred.

The **Off** value indicates that a component experienced a problem that is not transient (for example, the Recorder component stopped running). For such an alarm, the administrator must perform a corrective action to return the system to a functioning state.

- **Maximum Instances:** The maximum number of instances of the alarm that can be triggered (active) at one time.
- **Enabled:** This column indicates if the alarm is On or Off by default. If the alarm is On, the alarm can be triggered. If the alarm is Off, the alarm cannot be triggered. An administrator can configure this setting for each alarm.
- **Priority:** The numerical identifier of the default priority level of the alarm. These default priority levels are:

  0 - Information

  1 - Warning

  2 - Minor

  3 - Major

  These priority numbers and text values (Information, Warning, Minor, Major) are configurable by the administrator.
- **Alarm Summary:** A summary of the problem causing the alarm.
- **Alarm Details:** Provides detailed information about the alarm.
- **Corrective Action:** Explains how to correct or resolve the problem causing the alarm.
- **Advanced Corrective Action:** Describes additional corrective procedures if the initial attempts to correct the alarm fail.

  IMPORTANT Advanced corrective actions should be performed only by certified Verint personnel or certified partners of Verint. All other users should contact Verint support if the regular corrective actions fail to resolve the problem.

- **Cycle Multiplier -** The value used to determine how often to evaluate the rate counter to determine when to raise an alarm. The frequency with which the system evaluates the rate counter is based on the value of this attribute and the cycle time seconds that is configured globally for the alarm service. The formula for the frequency is:

  cycle multiplier * cycle time seconds

  Cycle time seconds is a non-configurable parameter that is set to 30 seconds. This element is not used with all alarms.
- **Object Name** - Lists the Windows Performance counter object name. This element is not used with all alarms.
- **Counter Name** - Lists the Windows Performance counter name (or other counter name if the counter is not a Windows Performance counter).
- **Trigger Rule** - Lists the condition on which an alarm is triggered. The possible values include:
  - **above** - The counter must be above the threshold to trigger the alarm.

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- **below** - The counter must be below the threshold to trigger the alarm.
- **notexist** - The alarm is raised if the counter does not exist. This trigger rule is useful for raising alarms about processes that are not running.

This element is not used with all alarms.

- **Trigger Type** - Lists the type of trigger for the alarm. The possible values include:
  - **peak** - The alarm is raised when the Windows performance counter goes above (if trigger rule is above) or below (if trigger rule is below) the configured threshold.

    The alarm is acknowledged by the system when the counter is above (if trigger rule is below) or below (if trigger rule is above) the acknowledge threshold.
  - **average** - The alarm is raised when the average value of the Windows performance counter goes above (if trigger rule is above) or below (if trigger rule is below) the configured threshold.

    The alarm is acknowledged by the system when the average value of the Windows performance counter is above (if trigger rule is below) or below (if trigger rule is above) the acknowledge threshold.

    The average is calculated over a period of time using this formula:

    cyclemultiplier * cycle time seconds.
  - **difference** - The alarm is raised when the Windows performance counter differs by a configured threshold and acknowledged when the counter differs less than the configured acknowledge threshold.

  This element is not used with all alarms.
- **Rate Counter -** The On value indicates the counter is a Windows Performance monitor rate counter. This element is not used with all alarms.
- **Threshold:** The threshold at which the alarm is triggered. Some alarms are triggered when the value rises above this threshold while other alarms are triggered when the value falls below this threshold. This value is configurable by the administrator. Not all of the alarms have thresholds.
- **Acknowledge Threshold:** Specifies the value at which the alarm is automatically acknowledged by the system. Generally, a certain amount of separation exists beneath this value and the **Threshold** value above.

  This separation exists to prevent alarms from being repeatedly triggered and acknowledged by the system when a value wavers back and forth across a specific threshold.

  For example, you can set an alarm's Threshold to above 3000 and the Acknowledge Threshold to below 2000. In this configuration, the alarm is triggered when the value exceeds 3000 and is not automatically acknowledged by the system until the value goes back below 2000.

  In this example, the alarm is not acknowledged until the value is safely below the threshold at which it was triggered. Setting the two values too close together can cause an alarm to be repeatedly triggered and acknowledged by the system if the system wavers back and forth across the threshold value.

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- **Trigger Delay** - Specifies the number of seconds that passes between the detection of an alarm and the time that the user is notified of the alarm (either through the appearance of the alarm in the user interface, an email notification, or an SNMP trap).

  This is an advanced setting that is needed only in particular circumstances. This setting prevents administrators from being notified of alarms that require no action by the administrator to resolve.

Advanced Corrective Actions Are For Certified Support Personnel Only

Advanced corrective actions for alarms should be performed only by certified Verint personnel or certified partners of Verint. All other users should contact Verint support if the regular corrective actions fail to resolve the problem causing the alarm.

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Chapter 5

Troubleshooting

This section discusses the following troubleshooting topics:

- Email Notifications Are Not Sent When an Alarm is Triggered, page 82
- Preventing the “Error Retrieving Performance Counter” Alarm in Mixed Language Server Configurations, page 82

Email Notifications Are Not Sent When an Alarm is Triggered

If emails are not sent when an alarm is triggered, verify that you have correctly configured the system to support email notifications.

The list below describes all configuration that must exist for an email to be sent when an alarm is triggered:

- The connectivity information for the SMTP server must be specified in **SMTP** settings on the **System Management > Enterprise Management > Settings** tab.
- The **Email Notification** setting must be selected in the **System Management > System Monitor > Global Alarm Settings** tab.
- The email addresses of the employees that you want to receive email notifications must be listed in at least one Notification Profile in the **System Management > System Monitor > Notification Profiles** tab.
- The Notification Profile above must be assigned to the alarm that is triggered. To assign Notification Profiles to alarms, see "Edit an Alarm" on page 37.

Preventing the "Error Retrieving Performance Counter" Alarm in Mixed Language Server Configurations

This section explains why the "Error Retrieving Performance Counter" alarm (number 30998) is raised with mixed language server configurations and how to prevent this alarm from being raised. This topic includes these sections:

- What is a Mixed Language Server Configuration?
- Windows Performance Counter and the Error Retrieving Performance Counter Alarm
- Preventing the Alarm from Being Raised

IMPORTANT The procedure in this section should be performed only by, or under the direct supervision of, an authorized Verint technical support representative.

What is a Mixed Language Server Configuration?

The problem described in this topic occurs only if you have mixed language server configurations. The term "mixed language server configuration" refers to either of the following configurations:

- A server with a Windows language pack installed that is different than the language of the Windows operating system (for example, a server with an English language operating system that has a Spanish language pack installed).
- A server with third-party software installed that is of a different language than the Windows operating system (for example, a Spanish language SQL server installed on the English language Windows operating system).

Windows Performance Counter and the Error Retrieving Performance Counter Alarm

The V11.x Alarmer component uses Windows operating system performance counters to raise and acknowledge alarms.

NOTE You can view these performance counter names, by running the **perfmon.exe** utility (available at <root>:\Windows\SysWOW64\). In the perfmon.exe utility, select **Data Collector Sets > System > System Performance** and then select **Performance Counters**.

For the performance counter to operate successfully with the V11.x Alarmer component, the performance counter names displayed in the perfmon.exe utility must match the V11.1 Alarmer component configuration.

A mixed language server configuration will cause performance counter names to display in two different languages: some in the language of the operating system and others in the language of the language pack or third party software.

When the Alarmer component is asked to query a Windows performance counter, and the counter name is not in the language expected by the Alarmer configuration, the Alarmer assumes the counter does not exist and raises the “Error Retrieving Performance Counter” alarm.

Preventing the Alarm from Being Raised

If you see an “Error Retrieving Performance Counter” alarm that results from a mixed server configuration, you can perform the following procedure to address this issue.

This procedure specifies the performance counter names it should translate and those it should not.

1. The “Error Retrieving Performance Counter” alarm will specify the display name of the alarm it could not query. Identify the alarm display name indicated by the “Error Retrieving Performance Counter” alarm.
2. Locate the alarm definition XML file that contains the alarm display name identified in step 1, as noted below.
   - All alarm XML definition files reside in the following directory on the server on which the Framework Applications server role is active: ***<installdirectory>*\Conf\alarm**
   - You must manually search through the alarm definition XML files in the ***<installdirectory>*\Conf\alarm** directory until you locate the alarm display name identified in step 1. (The name will be located inside of an <a:fullname><a:/fullname> element in one alarm definition within the file).
3. When you have located the single alarm XML definition file that contains the alarm identified in step 1, edit the file as follows:
   a. Locate the individual alarm definition of the alarm identified in step 1. The alarm definition is enclosed within the <componentalarm></componentalarm> tags.
   b. Within the alarm definition, make these two changes:

- In the <a:objectname translang="**On**">Process</a:objectname> tag, change the On value to **Off**.
- In the <a:countername translang="**On**">Private Bytes</a:countername> tag, change the On value to **Off**.

NOTE You can verify which counter language is being published by running the perfmon.exe utility and searching for the object that matches the <objectname> value and the counter name that matches the <countername> value.

4 At the top of the alarm definition XML file increment the version number.

- Locate the <systemmonitordefinition> tag at the very top of the file.
- Within the <systemmonitordefinition> tag, increment the version= value by 1. For example, if version="11001", change it to version="11002".

5 Save the updated alarm definition XML file. Wait for the Enterprise Manager application to update the AlarmConfig.xml file with the new information.

NOTE Never edit the AlarmConfig.xml file. This file is automatically regenerated and overwritten by the Enterprise Manager application.

6 From the Windows desktop, start the Windows Services Administrative Tool and restart the **Recorder Alarm Service**.

Restarting this service causes the Alarmer component to retranslate any necessary counter names.

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